Vowel Hiatus Resolution in Setswana

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Abstract

This study provides a descriptive account of vowel hiatus resolution (VHR) in Setswana, an agglutinative S31 Bantu language spoken in Botswana, South Africa, Zimbabwe, and Namibia (Bennett, Diemer, Kerford, Probert, & Wesi, 2016). Setswana has a CV/V syllable structure. Consecutive vowel sequences, known as vowel hiatus, can be found across syllable, morpheme, and word boundaries (Casali, 2011). Setswana disallows vowels in adjacent syllables (*V.V), as well as long vowels (Kadenge & Simango, 2014). Consequently, various vowel hiatus resolution strategies such as vowel elision, glide formation, diphthongization, and glottal stop epenthesis are employed to repair such sequences, resulting either in the re-syllabification of hetero-syllabic consecutive vowels into a single syllable, or in adjacent CV.CV syllables separated by a glottal stop. Elicited data from a native Setswana speaker, analyzed auditorily and acoustically (vowel formants, durations, and voicing), are used to analyze the different VHR strategies employed in various morpho-phonological contexts: word-internally at (1) morpheme boundaries and (2) syllable boundaries, and across (3) word boundaries. This study demonstrates that the type of hiatus resolution is dependent on two factors: the features of the vowels in combination, and the morpho-phonological context in which the hiatus sequence occurs. In other words, different morpho-phonological boundaries license or block different VHR strategies, resulting in variations in each context. At morpheme boundaries, hiatus sequences consisting of high and close-mid V1s have a tendency to glide when followed by a lower V2, whereas close-mid and open-mid V1s undergo elision when followed by a higher homorganic V2. By contrast, close-mid + high sequences, which undergo V1 elision at morpheme boundaries, are resolved using diphthongization at syllable boundaries. VHR across word boundaries is always resolved using glottal stop epenthesis regardless of the vowel combination.

1 Introduction

Vowel hiatus refers to the occurrence of consecutive vowel sequences in separate syllables (Casali, 2011). Hiatus sequences can be found in various morpho-phonological contexts, such as morpheme boundaries, syllable boundaries, and word boundaries. Languages vary in their treatment of hiatus sequences (Casali, 2021); they either allow it freely, allow it only in certain contexts, or disallow it completely (Kadenge & Simango, 2014). Languages that place constraints on vowel hiatus sequences use various strategies, known as vowel hiatus resolution (VHR), to eliminate them where they appear underlyingly (Casali, 2021). Some commonly used VHR strategies include vowel elision, glide formation, coalescence, diphthongization, and epenthesis (Casali, 2021). The nature of the hiatus resolution strategy employed can differ based on the morpho-phonological context and the features of the

vowels involved in the hiatus sequence (Kadenge & Simango, 2014; Vratsanos & Kadenge, 2017). Previous studies on vowel hiatus resolution have been conducted on many languages spanning different language families, including Ngiti (Kutsch Lojenga, 1994), Indonesian (Rosenthall, 1997), Obolo (Faraclas, 1982), Eastern Ojibwa (Howard, 1972), Kavalan (Lin, 2018), Korean (De Haas, 2010), and Greek (Baltazani, 2006). Major research on the general principles of vowel hiatus resolution that apply cross-linguistically has been done by Casali (1996; 1997; 2011; 2021), which provides a useful guideline for analysis of VHR in Setswana. This study contributes to the general typology of VHR in languages across the world.

Setswana (ISO 639-3: tsn), also known as Sechuana or Tswana, is the national language of the Republic of Botswana (Motzafi-Haller, Keuthmann, & Vossen, 2006), and an official language of Botswana alongside English (Tsonope, 1988). It is also widely spoken in South Africa, Zimbabwe, and Namibia (Bennett et al., 2016), with an estimated speaker population of 13,745,730 across all countries (Lewis, 2009). Setswana is an S31 Bantu language (Batibo, 2016), classified among the Sotho-Tswana (S30) group within the South-eastern zone of Bantu languages (Cole, 1955; Van de Velde, Bostoen, Nurse, & Philippson, 2019). Typical of Bantu languages, it has highly agglutinative morphology (Tsonope, 1988). Setswana's syllables generally follow a CV or V structure (Bennett et al., 2016). Crucially, the language allows V-initial stems, and its affixes, which are usually monosyllabic, have either CV or V structures, both vowel-final. Consequently, morpho-syntactic concatenation can result in vowel hiatus (CV-V) sequences both word-internally at morpheme and syllable boundaries, and across word boundaries (Kadenge & Simango, 2014). An example of vowel hiatus at prefix-stem morpheme boundaries in nouns is given below.[1]

(1) *m^{w}ìt^{h}ùdì*

mò-ìt^{h}ùdì
1-student
'student'

Setswana generally disallows such hiatus sequences and utilizes various VHR strategies such as vowel elision, glide formation, diphthongization, and glottal stop epenthesis. Interestingly, VHR is not merely a segmental phenomenon. Setswana is a tonal language with a privative tone system, with contrasts between marked (H)igh tone and inert (L)ow tone segments (Bennett et al., 2016; Chebanne, Creissels, & Nkhwa, 1997; Hyman, 2001). Upon the application of VHR, only one tone surfaces on the resulting syllable, and an H tone is never lost. While descriptions of Setswana's phonological systems and processes have previously appeared in the literature (Bennett et al.,

[1] The glossing convention used throughout the paper is as follows. The first line contains the surface representation transcribed in IPA; this line depicts the outcome of VHR on the underlying structure. The second line represents the underlying morpheme structure, which is separated by hyphens. The corresponding gloss is provided in the third line, followed by the English translation in the fourth line.

2016; Chebanne et al., 1997; Cole, 1955; Krüger & Pretorius, 2006), VHR is a robust process in Setswana that has yet to be studied in depth. As an active process affecting consecutive vowel sequences in Setswana, VHR necessitates systematic and careful analysis. To that end, this study provides an auditory and acoustic analysis of VHR in Setswana. The classification of VHR strategies is based on auditory judgments and spectrographic inspection, and is supported by acoustic measurements of vowel formants, vowel and gap durations, and voicing. The analysis is based on phonetic data collected from a native Botswanan speaker and contributes information on VHR to the phonetics and phonology of the language. It provides an analysis of VHR in three morpho-phonological contexts, namely

1. Morpheme boundaries, specifically prefix-stem boundaries in nominal *noun class-noun stem* (2-a) and verbal *subject agreement marker-verb stem* (2-b) constructions.

(2) a. *sìpònè*

sɛ̀-ìpònè
7-mirror
'mirror'

b. *báɔ́pélá dì-pìnà*

bá-ɔ̀pèlà dì-pìnà
3PL-sing.PRS-FV 8-song
'They sing songs.'

2. Syllable boundaries, between hetero-syllabic adjacent vowels (3-a) that do not coincide with morpheme boundaries.

(3) a. *pòunàmà*

pòùnàmà
9.lip
'lip'

3. Word boundaries, between noun-noun (4-b) and verb-noun (4-a) constructions.

(4) a. *kè-bɔ̀n-à^{ʔ}ɔ̀vɛ̀nɛ̀*

kè-bɔ̀nà ɔ̀vɛ̀nɛ̀
1SG-see.PRS-FV 9.oven
'I see an oven.'

b. *rè-fá mòsàdì^{ʔ}ɛ̀mɛ̀rɛ̀*

rè-f-á mò-sàdì ɛ̀mɛ̀rɛ̀
1PL-give.PRS-FV 1-woman 9.bucket
'We give the woman a bucket.'

This study argues that hiatus resolution in Setswana is driven by two syllable structure constraints:

(i) *V.V: two vowels may not occur as the nuclei of adjacent syllables.

(ii) *Vː: long vowels are not allowed.

Constraint (i) triggers VHR, and every VHR strategy described in this paper is a way of repairing a V.V sequence. V1 elision and glide formation leave a single vowel in a single CV syllable; diphthongization merges the two vowels into the nucleus of one syllable (CVV); and glottal stop epenthesis provides V2 with an onset, separating the vowels into adjacent CV.CV syllables. Constraint (ii) explains why identical vowels never merge into a long vowel: a sequence such as /a.a/ surfaces as a short [a] through V1 elision, without compensatory lengthening. Diphthongs are permitted because they consist of two different vowels, and are therefore not long vowels.

Furthermore, the VHR strategy employed depends on two factors:

1) The features of the two vowels in hiatus combination.

2) The morpho-phonological boundary in which the hiatus sequence appears.

In other words, while the features of the vowels in hiatus sequences play a role in determining the type of VHR applicable, the morpho-phonological context in which the hiatus sequence occurs presents additional constraints on the application of VHR, which override vowel-feature constraints and lead to varying profiles of VHR in the three morpho-phonological contexts. To elaborate, vowel hiatus at morpheme boundaries is resolved using V1 elision, glide formation, and diphthongization based on the features of the two vowels involved. VHR at syllable boundaries employs only glide formation, diphthongization, and the elision of identical vowels. The variation in VHR between the two contexts lies in the application of diphthongization. At syllable boundaries, any sequence involving a lower V1 and a higher V2 undergoes diphthongization. By contrast, at morpheme boundaries, diphthongization is restricted to hiatus sequences involving the low V1 vowel /a/ in combination with any other higher V2 vowel. Here, close-mid vowels /e̱ and o̱/ instead undergo elision when followed by a homorganic high V2. At word boundaries, on the other hand, vowel hiatus is always resolved using glottal stop epenthesis, regardless of the sequence of vowels involved. These patterns are summarized in Tables 7 and 8 (Section 4).

Cross-linguistic evidence suggests that numerous factors license or block the occurrence of vowel hiatus, such as “the nature of the prosodic or morpho-syntactic boundary at which hiatus arises, prominence of factors such as stress, vowel length and tone, minimal word length or weight conditions, the lexical or functional status of particular morphemes, rate of speech, and sensitivity to particular lexical items” (Casali, 2011). For example, vowel elision always targets V1 regardless of its position in the prefix or stem in chiShona; in ciCewa, however, it targets V1 in prefix positions but V2 in suffix positions (Casali, 1997; Kadenge & Simango, 2014). This study

on Setswana, therefore, contributes information on typological variations found in the factors that license or block VHR.

Furthermore, it is also typologically very common for languages to employ more than one VHR strategy. Bantu languages, which generally do not allow hetero-syllabic vowel hiatus sequences, use multiple strategies to resolve them (Vratsanos & Kadenge, 2017). For instance, to repair hiatus, ciNsenga employs glide formation, elision, and secondary articulation (Kadenge & Simango, 2014); chiShona employs five strategies: glide formation, secondary articulation, elision, vowel coalescence, and glide epenthesis (Mudzingwa & Kadenge, 2014). Interestingly, hiatus resolution operates in different morpho-syntactic and phonological contexts in both these languages (Kadenge & Simango, 2014). Zezuru employs glide formation, glide epenthesis, vowel elision, and vowel coalescence to resolve hiatus (Kadenge, 2010). Both Xitsonga (Vratsanos & Kadenge, 2017) and isiNdebele (Sabao, 2015) use glide formation, secondary articulation, elision, and coalescence. Lastly, Simbiti uses four strategies, namely vowel elision, diphthongization, glide formation, and glide epenthesis (Overton, 2018). The evidence from Setswana of multiple VHR strategies working in tandem contributes to the growing body of literature on VHR in other Bantu languages (ciNsenga and chiShona, Kadenge & Simango, 2014; Rutooro, Bickmore, 2021; Simbiti, Overton, 2018; Xitsonga, Vratsanos & Kadenge, 2017; Embosi, Rialland, Aborobongui, Adda-Decker, & Lamel, 2015), and provides an enriched understanding of factors that drive VHR in the language family.

This paper is structured as follows: Section 2 describes the methods of data collection and elicitation design. Section 3 provides preliminary information on Setswana's phonology, tone system, syllable structure, phonological processes, word classes, and word-formation processes. Section 4 consists of a detailed analysis of the different VHR strategies employed in the three different morpho-phonological contexts. This section also provides a brief insight into VHR at stem-suffix interfaces (Section 4.8). Section 5 discusses why the three contexts differ, and the gaps and exceptions in the data. Finally, Section 6 presents the conclusion of the study.

2 Method

For this project, 9 hours and 26 minutes of audio data focusing on VHR in Setswana were elicited. A Zoom H4n recorder, provided by the Linguistics Department at the University of Rochester, was the primary recording equipment. Recordings were annotated and inspected in Praat on macOS. The speaker's vowel plot was created from these formant measurements.

2.1 Elicitation design

All elicitations were designed using Creissels' (1996) extensive database of Setswana nouns and verbs and the online *Bab.la* Setswana dictionary (*Bab.la English-Tswana Dictionary*, n.d.). In every context, each V1 vowel was tested with examples of every other available V2 vowel. The database currently consists of 114 examples of VHR at morpheme boundaries, 33 examples of VHR at syllable boundaries, and 70 examples of VHR at word boundaries. See the Appendix for all examples.

To test for hiatus resolution at morpheme boundaries, 60 nouns with vowel-initial stems were tested against vowel-final noun class prefixes *ma-, mo-, lɛ-, sɛ-* and *di-*.

Similarly, verbs consisting of varied vowel-initial stems such as ***à**pàjà* 'cook', ***á**dìmà* 'lend', ***à**là* 'make bed', ***é**mà* 'stand', ***`ɛ**pà* 'dig', ***è**tà* 'visit', ***ò**pɛ̀là* 'sing', ***í**kàɲà* 'depend', and ***ú**tɬwà* 'hear' were tested with the vowel-final subject marker prefixes *k**è**-* 'I', *r**è**-* 'we', *b**á**-* 'they', ***ò**-* 'you', ***ó**-* 's/he', and the infinitive prefix *x**ò**-*.

Furthermore, to test for vowel hiatus resolution at word boundaries, numerous words with varied final vowels (*bɔ̀n**à*** 'see' and *rèkìl**è*** 'bought', *mòt^{h}**ù*** 'person', *mòsìmàn**è*** 'boy', *mòsàd**ì*** 'woman' and personal names *Theo, Isago,* and *Ubani*) were used in combination with nouns with varied initial vowels (***à**pòlɛ̀* 'apple', ***à**ɲànsì* 'onion', ***è**mɛ̀rɛ̀* 'bucket', ***è**nʗɛ̀nɛ̀* 'engine', ***ɔ̀**vènè* 'oven', ***ɔ̀**mfɔ̀lɔ̀pɔ̀* 'envelope', ***ò**fìsì* 'office', and ***I**sago* and ***U**bani*). These words, including both nouns and verbs, were then used to design simple sentence constructions that display the different V1-V2 hiatus combinations at verb-noun (5) and noun-noun (6) boundaries as shown below.

(5) *kè-bɔ̀n-a̲* *a̲pòlè*
1SG-see.PRS-FV 9.apple
'I see an apple.'

(6) *kè-f-á* *mò-s̀ìmànè̲* *ɔ̲̀mfɔ̀lɔ̀pɔ̀*
1SG-give.PRS-FV 1-boy 9.envelope
'I give the boy an envelope.'

Finally, 33 words (nouns, adjectives, and verbs) were selected from Creissels's (1996) database and *Bab.la* dictionary to test for vowel hiatus resolution at syllable boundaries.

During elicitation sessions, the speaker was asked to pronounce each example slowly, then at regular speed, tap out the number of syllables, and hum the tone. These tasks served as the diagnostics for syllabification. In slow speech, hiatus sequences were generally produced as two separate syllables, each with its own tone; VHR was observed at regular speed. A word-internal hiatus sequence was analyzed as resolved when, at regular speed, the speaker tapped one syllable fewer than in slow speech and only one of the two tones surfaced. Since L-L sequences surface as L either way, tone cannot serve as a diagnostic for them; for these sequences, the syllable count

was the primary evidence. Spectrograms were used to confirm these judgments: for each strategy, representative tokens were annotated in Praat to show the presence or absence of each vowel interval, glide transitions, and glottal closures (Figures 3–10 and 11). In addition to the formant measurements of the vowel inventory (Table 5), vowel and gap durations were measured for a small set of tokens (Section 4.7). No formant trajectories or F0 measurements of VHR outputs were taken. The classification of strategies is therefore based on auditory and spectrographic categorization, supported by these measurements; the figures serve as illustrative examples.

To analyze how VHR affects tone, vowel sequences with L-L, L-H, H-L, and H-H tone combinations were chosen. For instance, the high-tone subject agreement prefixes *bá-* 'they' and *ó-* 's/he', and the toneless subject agreement prefixes *kè-* 'I', *rè-* 'we', *ò-* 'you', and *xò-* were used in combination with various verb stems consisting of both high (*ádìmà* 'lend') and toneless initial vowels (*àpàjà* 'cook'). Similarly, the high-tone class 2 noun prefix *bá-* was tested along with multiple noun stems consisting of both high and toneless initial vowels.

2.2 Consultant Profile

All language data for this project was collected in collaboration with a native Setswana speaker from Phitshane Molopo, Botswana. The speaker is currently an undergraduate Chemical Engineering student at the University of Rochester, USA, and also speaks fluent English.

3 Phonological and Grammatical Preliminaries

This section provides a brief overview of all preliminary information required for a detailed comprehension of this study. It includes information on Setswana's phoneme inventory (Section 3.1), tone (Section 3.2), phonological processes (Section 3.3), syllable structure, and word classes and word-formation processes (Section 3.4).

3.1 Phoneme Inventory

The following is a description of the Setswana phoneme inventory based largely on data presented in Cole (1955) and Bennett et al. (2016), in addition to phonetic data elicited with the Botswanan speaker. Data collected during the project mostly converges with the information presented in Bennett et al. (2016). Minor variations are discussed below.

3.1.1 Consonants

Setswana, as spoken by the speaker, has 31 consonants, as shown in Table 1.

Table 1: **Consonant Inventory**

	Bilabial	**Labio-dental**	**Alveolar**	**Alveolar Lateral**	**Post-Alveolar**	**Palatal**	**Velar**	**Uvular**	**Glottal**
Plosive	p p^h b		t t^h d				k k^h		(ʔ)
Affricate			ts ts^h	tɬ $tɬ^h$	ʧ $ʧ^h$ ʤ			qχ	
Nasal	m		n			ɲ	ŋ		
Trill			r						
Fricative		(f)	s z		ʃ			χ	h
Approximant	w			l		j			

Setswana's bilabial and alveolar plosives display a three-way contrast: voiceless unaspirated, voiceless aspirated, and voiced. Voiceless unaspirated plosives and affricates can be realized as ejectives (Bennett et al., 2016). /f/ is known to alternate with /h/ in several dialects, as seen in: [k^h*ùtàfàtsà*] ⇄ [k^h*ùtsàhàtsà*] 'make (oneself) sad.' Additionally, /b/ varies with /ts/ in some words, such as [*màbɔ̀χɔ̀*] ⇄ [*màtsɔ̀χɔ̀*] 'arms'. The four nasal sounds /m/, /n/, /ɲ/, and /ŋ/ can be syllabic (Cole, 1955). Also, /d/, the voiced alveolar plosive, occurs only before the phonemic high vowels /i/ and /u/. It is treated as an allophone of /l/, because [l] never appears before /i/ or /u/ (Bennett et al., 2016).

A consonant word-list is given below in Table 2.

Table 2: **Consonant Word-list**

Phoneme	**Word (IPA)**	**Gloss**
/p/	pìnà	'song'
/t/	tàú	'lion'
/k/	kùbù	'hippopotamus'
/p^h/	p^hìrì	'hyena'
/t^h/	t^hàbà	'mountain'
/k^h/	k^hùdù	'tortoise'
/b/	bìnà	'dance'
/d/	dìʤɔ́	'food'

Table 2: **Consonant Word-list**

Phoneme	**Word (IPA)**	**Gloss**
[ʔ]	ʔáχá	‘build’
/ts/	tsɛ̀bɛ́	‘ear’
/ʧ/	sɛ̀ʧàbà	‘nation’
/tɬ/	tɬálà	‘famine’
/ts^{h}/	ts^{h}ɛ́χà	‘laugh’
/ʧ^{h}/	ʧ^{h}úswànì	‘ant’
/tɬ^{h}/	tɬ^{h}àpì	‘fish’
/ʤ/	ʤá	‘eat’
/qχ/	qχósì	‘king’
/f/	fùlà	‘graze’
/s/	sèhèlèlà	‘cut’
/z/	zú	*‘zoo’
/ʃ/	màʃì	‘milk’
/χ/	χóχà	‘pull’
/h/	hùmà	‘become rich’
/m/	mòt^{h}ù	‘person’
/n/	nàmà	‘meat’
/ɲ/	ɲàlà	*‘marry’
/ŋ/	ŋòŋòrɛ̀χà	‘complain’
/r/	rútà	‘teach’
/l/	lɛ́là	‘cry’
/w/	wà	‘fall’
/j/	jà	‘go’

Occasionally, the variety of Setswana documented here displays certain variations from what is documented in the literature. For instance, while Cole (1955) lists the velar fricative /x/, Bennett et al. (2016) identify the sound as the uvular /χ/. Based on the speaker’s assessment of the sound, the uvular fricative has been included in this consonant inventory. Also, the phoneme /z/ appears in the language as a result of language contact and borrowing,

as seen in *zù* 'zoo'; interestingly, this is documented in Cole (1955), but not in Bennett et al. (2016). Furthermore, although both Cole (1955) and Bennett et al. (2016) list *ʧúmà* 'hunt' as an example of the /ʧ/ phoneme, the speaker pronounces the word as *tsúmà* with a /ts/ instead. It was confirmed that the /ʧ/ sound does exist in the language, in words such as *sèʧàbà* 'nation'.

Lastly, although the glottal stop is non-phonemic, it can appear at the start of an underlying vowel-initial stem, or between a vowel-initial stem and a vowel-final prefix (Bennett et al., 2016), as in *χɔ̀-émà* → *χɔ̀-ʔémà* 'to stand'. This appears to be an instance of glottal stop epenthesis being used in VHR. However, in the provided context of vowel-final prefix and vowel-initial stem, none of the examples elicited in this documentation project yielded glottal stop epenthesis. This could be a variation between the Botswanan variety and South African variety of Setswana documented in Bennett et al. (2016). As explained in Section 4, VHR using glottal stop epenthesis is employed only at word boundaries in the Botswanan variety. The given hiatus sequence would be repaired with a glide, as in *χɔ̀-émà* → *χ*^{w}*émà*.

3.1.2 Vowels

Setswana has seven contrastive vowels—/i e̞ ɛ a ɔ o̞ u/ (Bennett et al., 2016), as shown in Figure 1. Converging with the data presented in Bennett et al. (2016), the close-mid vowels are slightly higher than [e] and [o], and are therefore transcribed as raised close-mid vowels /e̞/ and /o̞/.

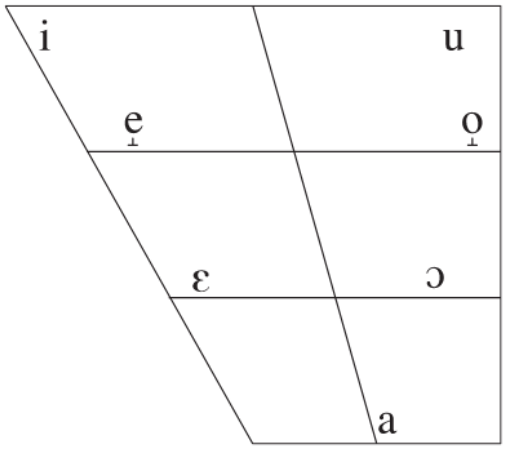


Figure 1: Vowel Chart (Bennett et al., 2016, p. 242)

The mid vowels, both open-mid (/ɛ ɔ/) and close-mid (/e̞ o̞/) have raised allophones [ɛ̝ ɔ̝] and [i̝ u̝], respectively (Bennett et al., 2016). According to the literature (Bennett et al., 2016; Cole, 1955), this raising is conditioned by the presence of a high vowel in the succeeding syllable, the locative suffix [-ŋ], or the consonants /s, ʃ, ts, ts^{h}, ʧ, ʧ^{h},

tɬ, tɬʰ, ɲ/. For the project's primary speaker, this raising was highly variable, appearing in certain contexts but not in others that warranted it. For example, *módùmò* 'noise' was produced as [mʊ́dùmò], with the close-mid vowel raised before the high vowel in the following syllable. Similar to the observations recorded in Bennett et al. (2016), the distinction between the raised allophones of the close-mid vowels /e̞ and o̞/ and the underlying high vowels /i and u/ is unclear for the speaker. There is also considerable overlap between the raised allophones of the open-mid vowels /ɛand ɔ/ and the close-mid vowels /e̞ and o̞/. The speaker described /ɛ/ and /ɔ/ as "a variation of" /e̞/ and /o̞/. A vowel word-list is given in Table 3.

Table 3: **Vowel Word-list**

Phoneme	**Word (IPA)**	**Gloss**
/i/	tsʰípì	'iron'
/e̞/	se̞χà	'cut'
/ɛ/	tsʰɛ́χà	'laugh'
/a/	tʰàbà	'mountain'
/ɔ/	sɔ̀kà	'twist'
/o̞/	ʧo̞mà	'hunt'
/u/	tsʰúbà	'set alight'

Throughout the analysis, vowels are grouped into the height and backness classes shown in Table 4. The terms *higher* and *lower* refer to the height scale high > close-mid > open-mid > low; since /a/ is the only low vowel, every other vowel counts as higher than /a/. The term *mid* covers both close-mid and open-mid vowels. Two vowels are *homorganic* when they share backness (both front or both back), and have *opposing backness* when one is front and the other back.

Table 4: **Vowel Classes Used in the Analysis**

Height	**Front**	**Central**	**Back**
High	i		u
Close-mid	e̞		o̞
Open-mid	ɛ		ɔ
Low		a	

A vowel plot of the speaker's formant values is given in Figure 2. Measurements were taken at the midpoint

of each vowel and averaged across three tokens for each example listed in Table 3. The mean formant values are given in Table 5.

Table 5: **Mean Formant Values (Hz) at Vowel Midpoint, Three Tokens per Vowel (SD in Parentheses)**

Vowel	Word	F1	F2	F3
i	*ts^{h}ípì*	377 (34)	2923 (147)	3428 (67)
e̞	*se̞χà*	437 (13)	2418 (117)	3143 (63)
ɛ	*ts^{h}ɛ́χà*	589 (19)	2434 (95)	3315 (145)
a	*t^{h}àbà*	811 (48)	1827 (27)	3375 (101)
ɔ	*sɔ̀kà*	593 (16)	1009 (78)	2972 (102)
o̞	*tso̞mà*	426 (18)	866 (15)	2757 (61)
u	*ts^{h}úbà*	363 (22)	1009 (52)	2729 (57)

The measurements support the seven-vowel analysis. The four mid vowels are clearly separated by F1: the close-mid vowels /e̞/ (437 Hz) and /o̞/ (426 Hz) lie much closer to the high vowels /i/ (377 Hz) and /u/ (363 Hz) than to the open-mid vowels /ɛ/ (589 Hz) and /ɔ/ (593 Hz). This is consistent with transcribing them as raised close-mid vowels, following Bennett et al. (2016). The relatively high F2 of /u/ (1009 Hz, above that of /o̞/) and of /a/ (1827 Hz) is likely due to the preceding coronal consonants in *ts^{h}úbà* and *t^{h}àbà*. Since these values come from a single word per vowel and three tokens each, they describe the speaker's citation forms rather than the full range of allophonic variation.

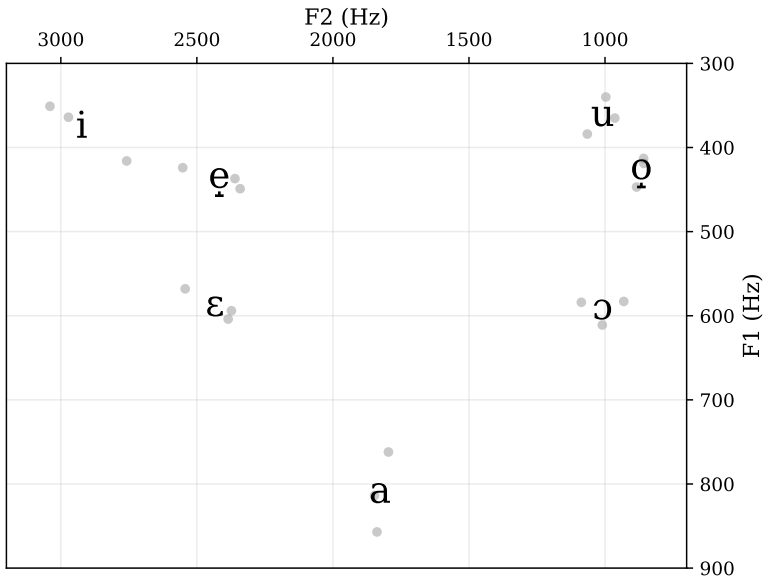


Figure 2: F1 and F2 of the Speaker's Vowels (Large Symbols: Means; Grey Dots: Individual Tokens)

3.2 Tone

Setswana is a tonal language with a privative tone system. It contrasts marked (H)igh and inert (L)ow segments (Chebanne et al., 1997; Hyman, 2011; Yip, 2002). The tone-bearing unit (TBU) is the syllable. Tone is lexically contrastive, as shown by the minimal pair *dìp^{h}írí* 'hyenas' (L-H-H) and *dìp^{h}ìrì* 'secrets' (L-L-L). Additionally, Setswana exhibits progressive High Tone Spreading (HTS) in certain conditions. For instance, if an underlyingly L-toned syllable follows an H-toned syllable, it may take on a surface H-tone (Chebanne et al., 1997).

Evidence of HTS can be found in subject agreement marking on verb stems. Example (7) provides a clear tonal contrast between L-L and H-L constructions. Here, the only differentiating factor between the second person singular subject agreement marker *ò-* and the third person singular subject agreement marker *ó-* is tone. When prefixed to an underlyingly L-toned verb stem *-apaja* 'cook', the H-tone on *ó-* spreads, resulting in surface high tones across the verb stem, as seen in (7-b). By contrast, the L-toned *ò-* does not bring about any surface level changes on the verb stem.

(7) a. *wàpàjà nàmà*

ò-àpàjà nàmà
2SG-cook-FV 9.meat
'You cook meat.'

b. *wápájá nàmà*

ó-àpàjà nàmà
3SG-cook-FV 9.meat
'S/he cooks meat.'

This study examines tone in the restricted context of vowel hiatus and seeks to answer a question that naturally arises: what happens to the tone on the vowel segments, once VHR has applied? Since the TBU is the syllable, vowels found in CV-V structures are marked for either H or L tone. This results in four possible tonal combinations: L-L, L-H, H-L, and H-H. As stated before, VHR is driven by the ban on vowels in adjacent syllables (*V.V); word-internally, it results in the re-syllabification of hetero-syllabic vowels into a single syllable. Since this syllable is a single TBU, only one of the two tones marked on the adjacent vowels surfaces after VHR. Any hiatus sequence that features an H-tone, either underlyingly or due to processes of High Tone Spreading (HTS), surfaces with an H-tone after VHR. HTS itself is a separate process from VHR. It is observed in verbs, where an H-toned prefix spreads H across the stem, but not in nouns, where only the merged syllable surfaces with H and the rest of the stem remains L (e.g., *bá-ètsì* → *báetsì* 'imitators').

Table 6 provides an overview of how tone surfaces after VHR.

Table 6: **Effects of VHR on Tone**

V1 Tone	**V2 Tone**	**Tone After VHR**	**Notes**
L	L	L	As the inert default tone, L does not create any surface level changes.
L	H	H	Only the H-tone on V2 surfaces.
H	L	H	The H of V1 is not lost when V1 elides or glides: it surfaces on the resulting syllable. In verbs, HTS then spreads H across the stem; in nouns, H does not spread to the rest of the stem.
H	H	H	Only the H tone on V2 surfaces.

The generalization behind Table 6 can be stated simply: *an H tone is never lost under word-internal VHR*. If either input vowel bears H, the resulting syllable surfaces with H; only L-L sequences surface as L. This includes cases where V1 is deleted: in *ó-ɔ̀lèlà* → *ɔ́lélá* 's/he picks up', the H of the elided prefix vowel survives on the following syllable. This pattern is often described as tone stability: the tone of a deleted vowel is retained and realized on a neighbouring syllable. HTS is a separate process, which applies after VHR in verbs.

3.3 Phonological Processes that Affect Vowels

Phonological processes are sensitive to morphological conditioning (Inkelas, 2014). They differ according to factors such as lexical class (noun or verb), morpheme type (stem or affix), and etymological origin (native words or loan words) (Sande, 2020). Cole (1955) describes the following four phonological processes that impact vowels:

Elision of vowels is observed in the noun class prefix *mo-*. The prefix's vowel /o̞/ is elided in contexts where it attaches to a noun stem beginning with /b/ or /f/. For example, **mo-b* → **mb-* → *mm- and *mo-f* → **mf-* → *mh-*. Additionally, in several dialects the vowels /e̞/, /ɛ/, and /o̞/ are elided when they are preceded and followed by the liquid consonant /l/, resulting in the /l/ becoming syllabic. For example,

(8) a. *mólèlò* 'fire' → *móllò*

b. *sèhèlèlà* 'cut' → *sèhèllà*

Coalescence is the fusion of two juxtaposed sounds to produce a new intermediate one.

CV1 + V2 → .CV3(:).[2]

In Setswana, this occurs as follows: a + i > e, a + u > o, and a + a > a. However, Cole (1955) points out that the process is relatively uncommon in Setswana, and is mainly found in the formation of locatives with the suffix *-(i)ŋ*. An example of this can be found in (9).

(9) *thaba* 'mountain' + *(i)ŋ* → *thabeŋ* 'locative of mountain'

Consonantalization refers to the process by which a vowel changes into a semi-vowel with consonantal function. The front vowels /i, e̯, and ɛ/ change to [j], and the back vowels /u, o̯, and ɔ/ change to [w]. Cole (1955) notes that this process is mainly found in the formation of nominal and adjectival diminutives through suffixation of *-ana/-ane*. He also notes that consonantalization of the stem-final vowel further triggers palatalization, velarization, or labialization of the preceding consonant. For example,

(10) a. *táù* 'lion' + *-ana* → *táwànà* 'lion cub'

b. *pódì* 'goat' + *-ane* → **pódìànè* → **pódyànè* → *pótsànè* 'baby goat.'

Vowel alternation refers to an interchange of the back vowels /u, o̯, and ɔ/ with [w] plus a front vowel /i, e̯, ɛ, and a/, as shown below (adapted from Cole, 1955, p. 50).

(11) a. Close *u* interchanges with *wi*; *ntsu* ⇌ *ntswi* 'eagle'

b. Close-mid *o̯* interchanges with *we̯*; *nole* ⇌ *nwele*, perfect stem of *-nwa* 'drink'

c. Open-mid ɔ interchanges with *wɛ*; *lenχole* ⇌ *lenχwele* 'knee'

In Setswana, phonological processes such as nasalization, palatalization, velarization, and assimilation can also affect consonants (Cole, 1955). Interestingly, palatalization and velarization, in particular, become relevant in the analysis of VHR at stem-suffix boundaries, where it is found that VHR through glide formation triggers a palatalization/velarization of the preceding consonant (see Section 4.8).

3.4 Word-formation Processes

Due to Setswana's agglutinative morphology, affixation is a very productive technique of word-formation (Krüger & Pretorius, 2006). Processes of prefixation and suffixation are used for both inflectional and derivational marking in the language (Cole, 1955). Affixes typically have CV or V syllable structure. Coupled with the availability

[2] The schematic representation presented here is based on Casali (1997). The parenthesized colon denotes optional compensatory lengthening. Additionally, periods are used to indicate syllable structure.

of vowel-initial stems, morpho-syntactic concatenation (both prefixation and suffixation) results in vowel hiatus sequences at several morpho-phonological boundaries (Kadenge & Simango, 2014).

3.4.1 Nouns

Setswana nouns consist of a nominal stem preceded by a noun class prefix (classes 1–14); the noun classes exhibit Bantu-typical systems of gender and agreement (Cole, 1955). As illustrated below, prefixation of the noun class morpheme to the nominal stem can result in hiatus sequences at the morpheme boundary.

(12) a. *mʷáχèdì̀*

mò-áχèdì̀
1-citizen
'citizen'

b. *sì̀.pà.tì̀*

sè-ì̀pàtì̀
7-excuse
'excuse'

Additionally, suffixes can be appended to the nominal stem to indicate augmentative or feminine (*-gadi/-adi*), diminutive (*-ana/-ane/-ɲana*), and locative (*-ŋ*) marking (Krüger & Pretorius, 2006). Vowel hiatus sequences are less common in stem-suffix structures. Nonetheless, they can be found in diminutive marking on nominal and adjectival forms. Diminutives have a variety of meanings, including small size or quantity, the young, or derogation and disparagement (Cole, 1955). Interestingly, unlike the majority of Bantu languages, Setswana uses suffixation for diminutive construction rather than prefixation, as shown below.

(13) *tá.wànà*

táù-ànà
lion-DIM
'lion cub.'

3.4.2 Verbs

The basic morphological structure of a Setswana verb requires either the infinitive morpheme *χo-* or agreement morphemes to be prefixed to the verb stem. Subject and object agreement marking, negation, and tense-aspect-mood inflections are also marked via prefixation of the morphemes to the verb stem (Krüger & Pretorius, 2006). Given below are some examples of hiatus sequences in infinitive marking (14) and subject agreement marking on a vowel-initial verb stem (15).

(14) *χú.tɬʷá*

χò-útɬʷ-á
INF-hear-FV
'To hear.'

(15) a. *kʲà.pà.jà nà.mà*

kè-àpàjà nàmà
1SG-cook-FV 9.meat
'I cook meat.'

b. *ó.pɛ́.lá dì̀.pí́.nà*

ó-òpɛ̀l-à dì̀-pí́nà
3SG-sing-FV 8-song
'S/he sings songs.'

A rich variety of verbal extensions are also formed using the process of suffixation (Cole, 1955). These suffixes influence a verb's lexical meaning and its syntactic combinability (Krüger & Pretorius, 2006). Setswana's most prominent verb extensions include passives (*-wa or -iwa*), neuters (*-exa, -ala, -esexa,* etc.), applicatives (*-ela, -etsa*), causatives (*-isa, -ya*), extensives (*-aka*), reciprocals (*-ana*), and reversives (*-ola*). Furthermore, a primitive verb stem usually has a terminative vowel *-a* (glossed FV). However, because this terminative vowel is substituted by various suffixal elements during conjugation (Cole, 1955; Krüger & Pretorius, 2006), vowel hiatus sequences at stem-suffix boundaries in verbal constructions are uncommon.

4 Analysis

This section analyzes the various VHR strategies used in Setswana. Each morpho-phonological context licenses distinct applications of VHR. For example, morpheme boundaries warrant the usage of vowel elision, glide formation, and diphthongization. However, close-mid + high sequences, which elide at morpheme boundaries, undergo diphthongization at syllable boundaries. Furthermore, vowel hiatus is always repaired using glottal stop epenthesis at word boundaries. The open-mid vowel /ɛ/ displays an interesting optionality in its choice of VHR strategy, undergoing either vowel elision or glide formation at morpheme boundaries (Section 4.3).

Two generalizations hold across all strategies. First, the target of VHR is always V1: it is V1 that elides or glides, and at morpheme boundaries, V1 is always the vowel of the prefix. V2, the initial vowel of the stem, is never deleted. Second, word-internal VHR is a regular-speed phenomenon: in slow speech, hiatus sequences are generally produced as two syllables (see Section 2).

Tables 7 and 8 summarize the strategy observed for each V1-V2 combination at morpheme and syllable boundaries. At word boundaries, every combination is resolved by glottal stop epenthesis.

Table 7: **VHR at Morpheme Boundaries (prefix V1 × stem V2)**

V1 \ V2	**i**	**e̙**	**ɛ**	**a**	**ɔ**	**o̙**	**u**
i	E	G	–	G	–	G	–
e̙	E	E	G	G	G	–	G
ɛ	E	E	E	G/E[a]	G/E	–	E
a	D	D	D	E	D	D	D
o̙	G	G	G	G	E[b]	E	E

E = V1 elision; G = glide formation; D = diphthongization; – = not tested. [a]Optional in verbs; nouns show glide formation only. [b]/o̙/ + /ɔ/ surfaces as [ɔ], rather than with the glide predicted by the general rule.

Table 8: **VHR at Syllable Boundaries (root-internal V1 × V2)**

V1 \ V2	**i**	**e̙**	**ɛ**	**a**	**ɔ**	**o̙**	**u**
i	E	G	–	G	–	G	–
e̙	D	–	–	–	G	–	G
ɛ	–	–	E	–	D	–	–
a	D	D	D	E	–	D	D
o̙	D[c]	G	–	G	–	E	D
u	G	G	–	G	–	–	E

E = V1 elision; G = glide formation; D = diphthongization; – = not tested. [c]Except *bòìdììdì* 'multitudes', which shows glide formation.

Why the three contexts license different strategies is discussed in Section 5.

4.1 Vowel Elision

Vowel elision refers to the deletion of one of the two vowels, with or without compensatory lengthening.[3]

CV1 + V2 → .CV1/2(:).

In Setswana, a mid V1 vowel (close-mid /e̙ o̙/ or open-mid /ɛ/) undergoes elision when it is followed by a higher homorganic V2 vowel. The open-mid /ɛ/ also elides before /u/, and optionally before /a/ and /ɔ/ (Section 4.3). Identical vowels likewise undergo V1 elision (Section 4.5). No compensatory lengthening was perceived.

Therefore, a front, close-mid vowel /e̙/ is elided when followed by the front, high vowel /i/. Similarly, the back, close-mid vowel /o̙/ undergoes elision when followed by a back, high vowel /u/.

$\mathbf{V_{MID} \rightarrow \emptyset \,/\, _\ V_{HIGHER,\ HOMORGANIC}}$

[3] Although V1 elision is cross-linguistically far more common, attested in languages such as Yoruba, there are languages like Etsako that display V2 elision (Casali, 1997).

$\mathbf{V_{FRONT\text{-}MID} \rightarrow \emptyset \; / \; _ \; V_{FRONT,\ HIGHER}}$

$\mathbf{V_{BACK\text{-}MID} \rightarrow \emptyset \; / \; _ \; V_{BACK,\ HIGHER}}$

Upon elision of V1, only one tone surfaces on the remaining syllable, and an H tone is retained (Section 3.2). Therefore, L-L tone sequences naturally surface as L; L-H and H-H sequences surface as H; and H-L sequences, as a result of HTS, also surface as H.

4.1.1 Morpheme Boundaries

Vowel elision occurs only at morpheme boundaries. The use of this strategy is not found at syllable or word boundaries. The following are examples of V1 elision at prefix-stem boundaries in noun class-noun stem (16) and subject agreement marker-verb stem (17) constructions.

(16) a. *sì̀.pà.tì̀*

sè-ì̀pàtì̀
7-excuse
'excuse'

b. *mú.wà.nè*

mò-úwànè
3-mist
'mist'

(17) a. *kí́.kà.ɲà bò.nè*

kè-í́kàɲ-à bònè
1SG-depend.on.PRS-FV them
'I depend on them.'

b. *χú.tɬwà*

χò-útɬwà
INF-hear-FV
'To hear.'

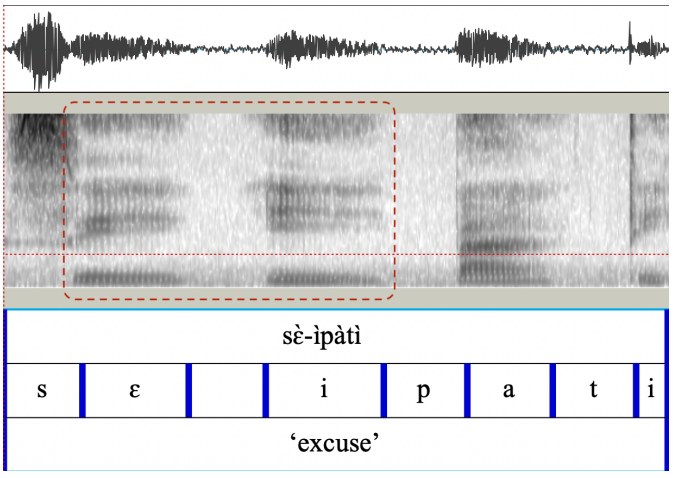


Figure 3: Morpheme Boundary - Before V1 Elision. *sè-ìpàtì* 'excuse' in slow speech. The prefix vowel [ɛ] and the stem vowel [i] are produced as two separate vowel intervals, separated by a short pause, each forming its own syllable.

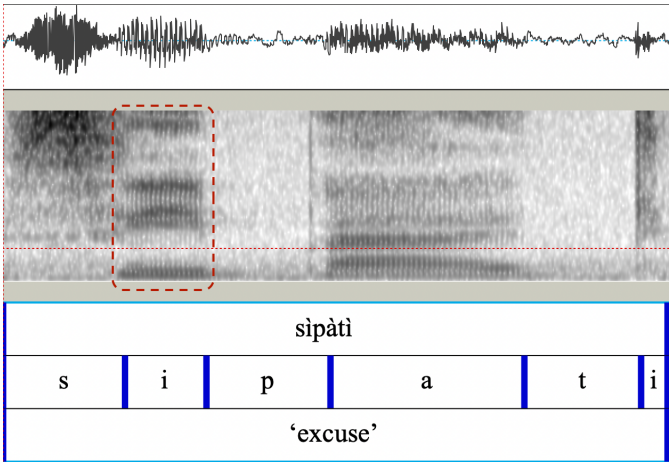


Figure 4: Morpheme Boundary - After V1 Elision. *sìpàtì* 'excuse' at regular speed. [s] is followed directly by [i]: the [ɛ] interval is absent, and [i] is not lengthened.

4.2 Glide Formation

Glide formation refers to the conversion of V1 into a semivowel, as formalized below.[4]

CV1 + V2 → .CGV2(:).

In Setswana, vowel hiatus is repaired using glide formation in sequences where V1, a high or close-mid vowel /i u and e̯ o̯/, is followed by a homorganic lower vowel or a vowel with opposing backness. Front vowels convert into the palatal [j], while back vowels labialize into [w]. No compensatory lengthening was perceived.

$V_{HIGH/CLOSE\text{-}MID}$ → G / _ $V_{LOW/OPEN\text{-}MID}$

i) $V_{FRONT,\ HIGH/CLOSE\text{-}MID}$ → j / _ $V_{FRONT,\ LOW/OPEN\text{-}MID;\ BACK}$

ii) $V_{BACK,\ HIGH/CLOSE\text{-}MID}$ → w / _ $V_{BACK,\ LOW/OPEN\text{-}MID;\ FRONT}$

Again, once V1 glides, the two vowels form a single syllable with only one surface tone, and an H tone is retained. Similar to V1 elision, L-L tone sequences surface as L; L-H and H-H sequences surface as H; and H-L sequences, as a result of HTS, also surface as H.

4.2.1 Morpheme Boundaries

The following are examples of glide formation at prefix-stem boundaries in noun class-noun stem (18) and subject agreement marker-verb stem (19) constructions.

[4] Cross-linguistically, languages can limit the availability of glide formation based on vowel backness and height. For instance, in Xhosa, only back round vowels glide, while in Polish only front vowels glide. Glide formation is more common when V1 is high, but Chicano Spanish has attested glide formation of mid V1s (Casali, 1995, 2011; Rosenthall, 1997). There are also languages, like Ganda, which display glide formation of both high-front and high-back V1 vowels (Casali, 2011; Rosenthall, 1997).

(18) a. *dʲàpàrò*
dì-àpàrò
8-cloth
'clothes'

b. *mʷàxì*
mò-àxì
1-builder
'builder'

c. *mʲú.wà.nè*
mè-úwànè
4-mist
'mists'

(19) a. *kʲá.dì.mà bù.kà*
kè-ádìm-à bùkà
1SG-lend.PRS-FV
9.book
'I lend a book.'

b. *wɛ́.pá ǹmù*
ó-èp-à
3SG-dig.PRS-FV
ǹmù
3.soil
'S/he digs soil.'

c. *kʲɔ̀.pè.là dì.pì.nà*
kè-ɔ̀pèl-à
1SG-sing.PRS-FV
dì-pìnà
8-song
'I sing songs.'

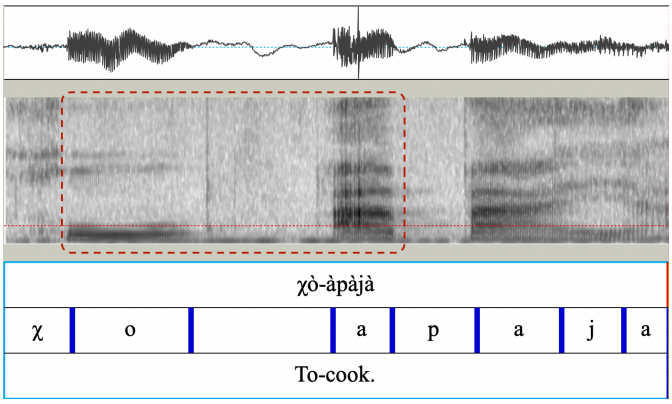


Figure 5: Morpheme Boundary - Before Glide Formation. *χò-àpàjà* 'to cook' in slow speech. [o̯] and [a] are produced as two separate vowel intervals, separated by a short pause.

4.2.2 Syllable Boundaries

Glide formation is also found word-internally at syllable boundaries, in both loanwords (e.g., *museum*, *radiator*) and native words (e.g., *lè-sʲèlà* 'orphan', from /lè-sì.è.là/; *bùìpùélètsò* 'repetition'; *kùkùànà* 'gathering').

(20) a. *mʲú.sʲá.mù*
mì.ú.sì.á.mù
9.museum
'museum'

b. *rá.dʲè.tà.rà*
rà.dì.e.tà.rá
9.radiator
'radiator'

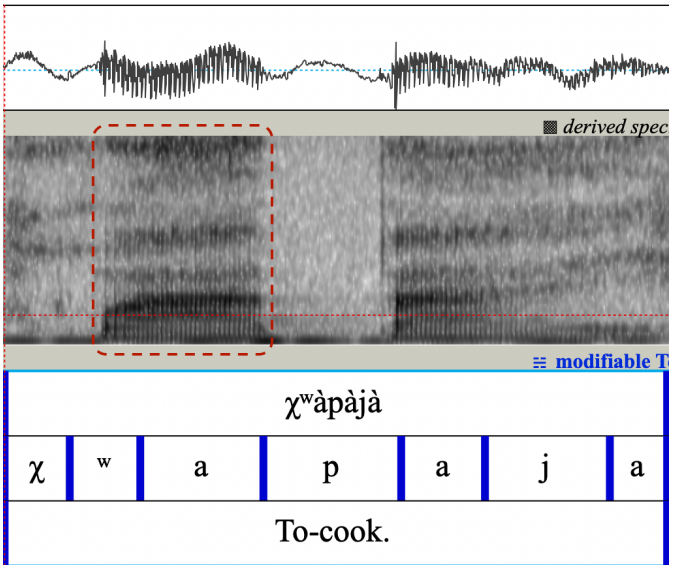


Figure 6: Morpheme Boundary - After Glide Formation. *χʷàpàjà* 'to cook' at regular speed. The [o̯] interval is replaced by a brief labial transition into [a], with low F2 at the onset of the vowel.

c. *bʷì.pʷé.lè.tsò*

bù.ì.pù.é.lè.tsò
2a-repetition
'repetition'

d. *kù.kʷà.nà*

kù.kù.à.nà
9.gathering
'gathering'

4.3 The Case of /ɛ/

Interestingly, the open-mid V1 vowel /ɛ/ displays optionality in the choice of VHR strategy: for the same item, the speaker produced both an elided and a glided form. This optionality is found with V2 /ɔ/ in both nouns and verbs, and with V2 /a/ in verbs only. In nouns, /ɛ/ + /a/ was always resolved by glide formation, and /ɛ/ + /u/ was always resolved by elision. Both variants were produced at regular speed for the same items; the choice between them is in free variation and does not depend on speech rate. The same item can also vary across sessions: *rɛ̀-ɔ̀lèlà* 'we pick up' was produced with elision ([rɔ̀lèlà]) in one session and with glide formation ([rʲɔ̀lèlà]) in another.

$V_{\text{FRONT, OPEN-MID}} \rightarrow$ Ø / _ $V_{\text{LOW/BACK OPEN-MID}}$

OR

$V_{\text{FRONT, OPEN-MID}} \rightarrow$ G / _ $V_{\text{LOW/BACK OPEN-MID}}$

This can be contrasted with the front close-mid vowel /e̯/, which elides when followed by a higher V2 (4.1), and glides when followed by a lower V2 (4.2).

Following are some examples of optional VHR displayed at prefix-stem boundaries, both noun class-noun stem, as seen in (21) and (22), and subject agreement marker-verb stem, as in (23) and (24). V1 elision is found in

(21-a), (22-a), (23-a), and (24-a), while (21-b), (22-b), (23-b), and (24-b) reflect glide formation.

(21) a. *lɔ́bò*

lè-ɔ́bò
5-fence
'fence'

b. *lʲɔ́bò*

lè-ɔ́bò
5-fence
'fence'

(22) a. *sɔ́pò*

sè-ɔ́pò
7-group
'group'

b. *sʲɔ́pò*

sè-ɔ́pò
7-group
'group'

(23) a. *ràpàjà nàmà*

rè-àpàjà nàmà
1PL-cook.PRS-FV 9.meat
'We cook meat.'

b. *rʲàpàjà nàmà*

rè-àpàjà nàmà
1PL-cook.PRS-FV 9.meat
'We cook meat.'

(24) a. *rɔ̀lèlà m̀mù*

rè-ɔ̀lèlà m̀mù
1PL-pick.up.PRS-FV 3.dirt
'We pick up dirt.'

b. *rʲɔ̀lèlà m̀mù*

rè-ɔ̀lèlà m̀mù
1PL-pick.up.PRS-FV 3.dirt
'We pick up dirt.'

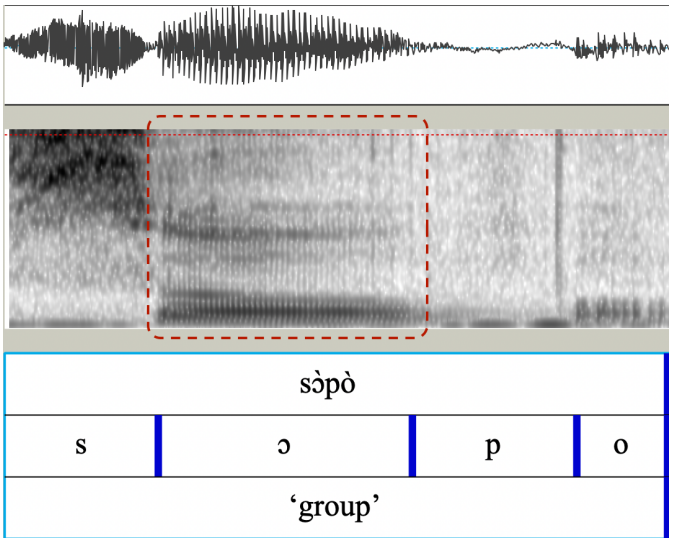


Figure 7: /ɛ/ - Resolved Using V1 Elision. *sɔ́pò* 'group'. [s] is followed directly by [ɔ], with no trace of the prefix vowel [ɛ].

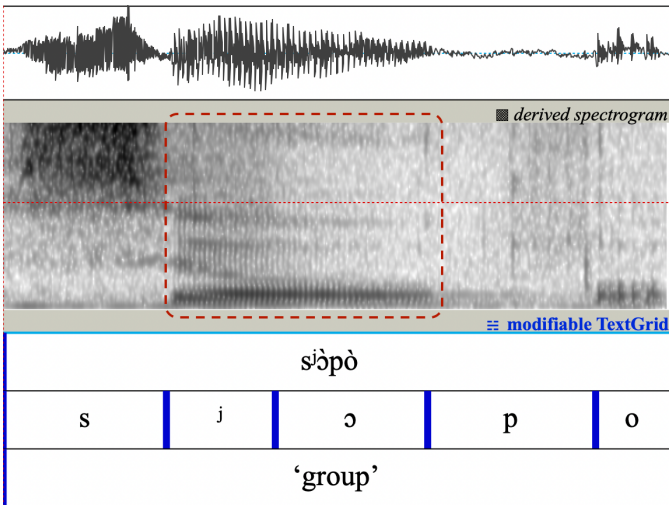


Figure 8: /ɛ/ - Resolved Using Glide Formation. *sʲɔ́pò* 'group'. The prefix vowel surfaces as a brief palatal transition, with high F2, between [s] and [ɔ].

Tone sequences are resolved similarly to vowel elision and glide formation.

4.4 Diphthongization

Diphthongization refers to the syllabification of the two vowels into the nucleus of a single syllable.

CV1 + V2 → .CV1V2.

The patterns of diphthongization appear to be slightly different at morpheme and syllable boundaries.

4.4.1 Morpheme Boundaries

Diphthongization occurs at morpheme boundaries exclusively when the hiatus sequence involves the low V1 vowel /a/ in combination with any higher V2 vowel (/i, u, ɛ, ɔ, e̯, and o̯/), as shown below.

.V_{LOW}.$V_{HIGH/MID}$. → .$V_{LOW}V_{HIGH/MID}$.

Evidence of diphthongization can be found in the difference in how tone is marked on such constructions in slow and regular speech. In slow speech, hiatused vowels are marked with distinct tones, indicating that they are hetero-syllabic. In regular speech, by contrast, both vowels are marked with a single tone suggesting re-syllabification into one syllable. Converging with tonal patterns observed with vowel elision and glide formation, L-L sequences surface as L, and L-H, H-L, and H-H sequences surface as H. Interestingly, diphthongs do not result in contour tones, at least as observed within the limited dataset of this project.

Below are some examples of diphthongization of hiatus sequences at prefix-stem boundaries in noun class-noun stem (25) and subject agreement marker-verb stem (26) constructions.

(25) a. *mái.kà.tɬò*

mà-í.kà.tɬò
6-regret
'regrets'

b. *bàɛ.mì.sì*

bà-ɛ̀mìsì̀
2-bridesmaid
'bridesmaids'

(26) a. *báo.pɛ́.lá dì.pí.nà*

bá-ò.pɛ̀.l-à dì-pí.nà
3PL-sing.PRS-FV 8-song
'They sing songs.'

b. *bái.ká.ɲá rò.nà*

bá-íkàɲ-à rònà
3PL-depend.PRS-FV us
'They depend on us.'

4.4.2 Syllable Boundaries

On the other hand, at syllable boundaries, every hiatus sequence involving a lower V1 vowel and a higher V2 vowel undergoes diphthongization, as schematized below.

.V$_{\text{LOW/MID}}$.V$_{\text{MID/HIGH}}$. → **.V$_{\text{LOW/MID}}$V$_{\text{MID/HIGH}}$.**

Interestingly, vowel sequences consisting of close-mid V1 vowels (/e̯ and o̯/) and high V2 vowels /i and u/ that undergo elision at morpheme boundaries (Section 4.1) are diphthongized at syllable boundaries, as seen in (27-c) and (27-d).

(27) a. *m̀mài.lè*

m̀mà.ì.lè
9.mile
'mile'

c. *sóu.sò*

só.ù.sò
9.sauce
'sauce'

b. *sè.kàu.sù*

sè-kà.ù.sù
7-sock
'sock'

d. *t^{h}èi.pì*

t^{h}è.ì.pì
9.tape
'tape'

Many of the syllable-boundary items in the dataset are loanwords (e.g., *mile*, *sauce*, *tape*). Diphthongization at syllable boundaries could therefore reflect the adaptation of borrowed vowel sequences, rather than the syllable-boundary context itself. However, native words show the same pattern, as seen in (28).

(28) a. *mò.lào*

mò-là.ò
3-law
'law'

b. *lào.lè.là*

là.ò.lè.là
allow
'allow'

c. *bú.lòi*

bú.lò.ì
witchcraft
'witchcraft'

d. *pòu.nà.mà*

pò.ù.nà.mà
9.lip
'lip'

4.5 V1 = V2: Elision of Identical Vowels

In instances where the vowel hiatus sequence consists of two identical vowels, V1 elides, and the sequence surfaces as a single short vowel. This is a special case of V1 elision (Section 4.1): since *V: rules out merging the two vowels into a long vowel, deletion is the only way to satisfy *V.V. No compensatory lengthening was perceived.

Here again, only one tone surfaces, following the same patterns as observed above.

4.5.1 Morpheme Boundaries

Given below are examples of prefix-stem constructions with identical consecutive vowels, noun class-noun stem (29) and subject agreement marker-verb stem (30).

(29) a. *sè.lè*

sè-èlè
7-sentence
'sentence'

b. *má.nò*

mà-ánò
6-plan
'plans'

(30) a. *xò.pè.là*

xò-òpèl-à
INF-sing-FV
'To sing.'

b. *bá.dí.má pè.nè*

bá-ádì̀m-à pènè
3PL-borrow.PRS-FV 9.pen
'They borrow a pen.'

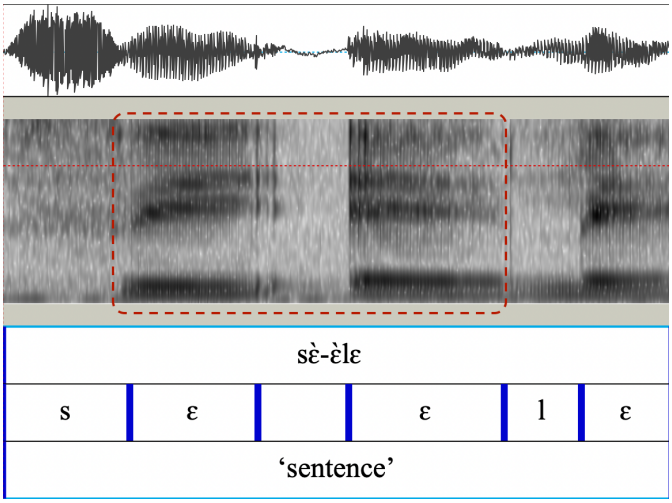


Figure 9: Morpheme Boundary with V1 = V2, Pronounced in Slow Speed. *sè-èlè* 'sentence' in slow speech. The two identical vowels are produced as two separate syllables, separated by a short pause.

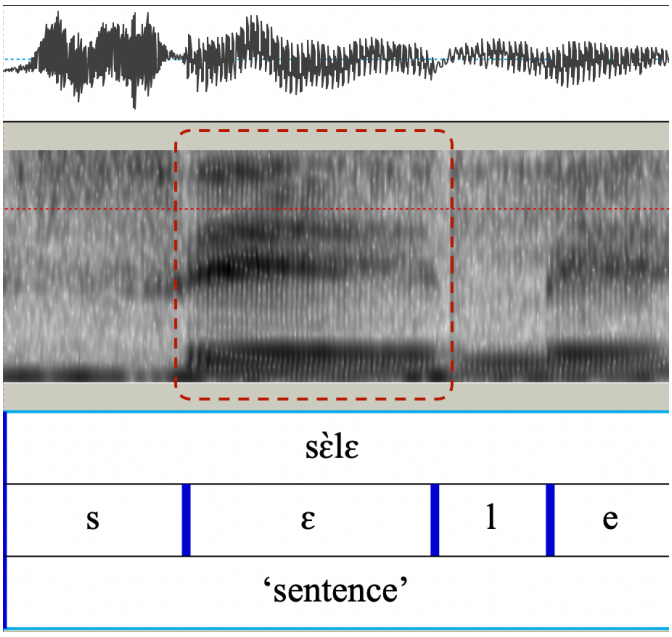


Figure 10: Morpheme Boundary with V1 = V2, Pronounced in Regular Speed. *sèlè* 'sentence' at regular speed. Only a single [ɛ] interval remains, with no compensatory lengthening.

4.5.2 Syllable Boundaries

Elision of identical vowels is also found at syllable boundaries.

(31) a. *sà.tà.nì*

sà.à.tà.nì
9.Satan
'Satan'

b. *nè.lò*

nè.è.lò
9.offer
'offer'

c. *lè.tɬò*

lè.tɬò.ò
5-hate
'hate'

d. *sè.kù.pù*

sè-kù.ù.pù
7-bottle
'bottle'

4.6 Glottal Stop Epenthesis

Epenthesis refers to the insertion of a consonant between V1 and V2 to interrupt hiatus. This consonant can be a glottal stop [ʔ], a homorganic glide that shares the same roundness as V1 or V2, a fricative [h], or a coronal consonant, such as a rhotic or [t].

CV1 + V2 → CV1.CV2. or CV1.GV2. or CV1.ʔV2.

In Setswana, vowel hiatus is always repaired using glottal stop epenthesis at word boundaries, regardless of the sequence of vowels involved.

Ø → ʔ/ V# _ #V

This is a case of phonological epenthesis: the inserted glottal stop provides an onset for V2, so the two vowels remain in separate syllables (CV.ʔV). Because no re-syllabification takes place, both vowels keep their own tones.

4.6.1 Word Boundaries

Given below are a few examples showing vowel hiatus resolution at word boundaries between verb-noun (32-a), (32-b) and noun-noun (32-c), (32-d) constructions.

(32) a. *kè-bɔ̀n-àʔàpòlè*

kè-bɔ̀nà àpòlè
1SG-see.PRS-FV 9.apple
'I see an apple.'

b. *kè-rɛ́kɨ̀lèʔɔvɛ̀nè*

kè-rɛ́kɨ̀lè ɔvɛ̀nè
1SG-buy-PST-FV 9.oven
'I bought an oven.'

c. *kè-fà mòsì̀mànè*^{ʔ}*èmèrè*

kè-f-à mò-sì̀mànè èmèrè
1SG-give.PRS-FV 1-boy 9.bucket
'I give the boy a bucket.'

d. *kè.fì̀lè mòsàdì̀*^{ʔ} *ɔ̀mfɔ̀lɔ̀pɔ̀*

kè-fì̀l-è mò-sàdì̀ ɔ̀mfɔ̀lɔ̀pɔ̀
1SG-give-PST-FV 1-woman 9.envelope
'I gave the woman an envelope.'

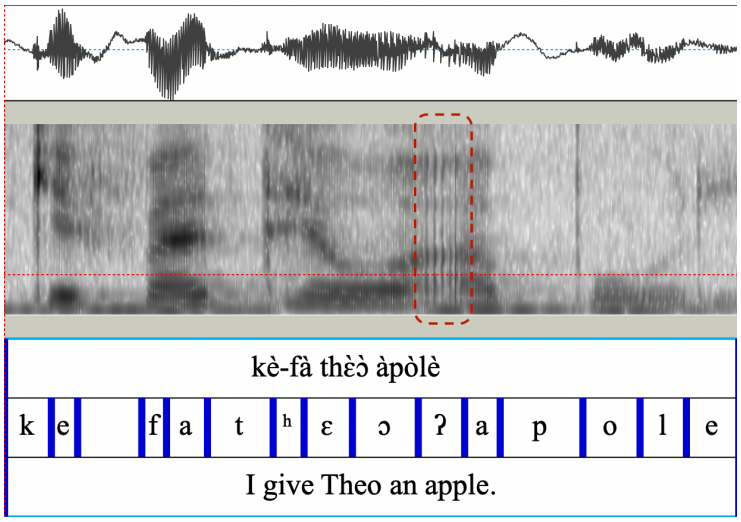


Figure 11: Glottal Stop Epenthesis. *kè-fà t*^{h}*èɔ*^{ʔ}*àpòlè* 'I give Theo an apple'. A glottal stop, marked as [ʔ] in the TextGrid, separates the final [ɔ] of *Theo* from the initial [a] of *apole*.

4.7 Duration and Voicing Measurements

To supplement the auditory and spectrographic judgments, durations were measured for a small set of recordings in which the same item was produced both slowly and at regular speed. For *sè-ìpàtì*, the values come from manual segmentation in Praat. For the other items, vowel intervals were identified from voicing and intensity in 5 ms analysis frames, so values are accurate to approximately ±10 ms. Each value comes from a single token. The results are given in Table 9.

Two observations follow. First, in slow speech, V1 and V2 are always separated by a silent interval of 165–310 ms, confirming that the two vowels are produced as separate syllables. At regular speed, voicing is continuous from the onset consonant through the output vowel. Second, the output vowel at regular speed (80–245 ms) is no longer than a single vowel in slow speech (144–360 ms), and much shorter than the combined duration of V1 and

Table 9: **Vowel and Gap Durations (ms) in Slow and Regular Speech**

Item	Strategy	Slow V1	Slow gap	Slow V2	Regular V
sè-ìpàtì → *sìpàtì*[a]	V1 elision	320	235	360	80
sè-èlè → *sèlè*	V1 = V2	195	165	220	245
rè-ádìmà → *rádìmà*	V1 elision (/ɛ/)	310	310	145	220
χò-àpàjà → *χʷàpàjà*	Glide formation	280	300	144	233

[a]Hand-segmented in Praat (the tokens shown in Figures 3 and 4).

V2 (415–680 ms). In *sìpàtì*, the output [i] (80 ms) is also shorter than the following [a] (190 ms) in the same word, which is expected if [i] is a single short vowel: high vowels are intrinsically shorter than low vowels, and [a] is in the penultimate syllable. This supports the claim that V1 is deleted rather than merged with V2 into a long vowel, and that no compensatory lengthening takes place. Since slow and regular productions differ in speech rate, and the vowel of *sèlè* is in the penultimate syllable, which may be lengthened, these comparisons should nonetheless be treated with caution.

Glottal stop epenthesis at word boundaries can also be observed in the recording of *rádìmàʔàŋànsì* 'We borrow an onion'. At regular speed, the boundary between *rádìmà* and *àŋànsì* is marked by approximately 50–60 ms of irregular, widely spaced glottal pulses (creaky voice), rather than by a full closure. In slow speech, the two words are separated by a pause of about 235 ms.

4.8 What happens at stem-suffix boundaries?

The analysis in this section is preliminary, as it is based on a small number of diminutive forms. Due to the primarily substitutive nature of suffixation in Setswana (Krüger & Pretorius, 2006), vowel hiatus sequences are rare in stem-suffix contexts. However, limited examples of such sequences can be found in diminutive marking in the language, where the morpheme *-ana* suffixes to a noun or adjective stem. Therefore, the following analysis of VHR at stem-suffix boundaries is restricted to the interaction of noun stems consisting of varied final vowels with the diminutive suffix *-ana*, and thus focuses primarily on V1-a constructions.

VHR at stem-suffix boundaries appears to be different from VHR at prefix-stem boundaries analyzed above, at least in their surface realizations. In other words, despite the hiatus sequence comprising the same combination of vowels, the resulting surface forms at prefix-stem and stem-suffix boundaries appear to be different, as shown below.

Hiatus sequences consisting of a high /i u/ or close-mid /e̞ o̞/ V1 vowel followed by a low vowel tend to undergo glide formation, as demonstrated in Section 4.2. However, diminutive suffixation results in the following

surface forms.

(33) a. *kòlòɕʒʷànà*

kòlòbò-ànà
9.pig-DIM
'piglet'

b. *lòlèŋʷànà*

lòlèmè-ànà
9.tongue-DIM
'small tongue'

(34) a. *tsʰìpi̗tsʷànà*

tsʰìpi̗-ànà
9.metal-DIM
'small piece of metal'

b. *lègòtʃʷànà*

lè-gòdù-ànà
5-thief-DIM
'petty thief'

Instead of consistently being resolved using glide formation as seen in prefix-stem constructions, the diminutives in (33) and (34) surface with *ɕʒʷà*, *ŋʷà*, *tsʷà*, and *tʃʷà*, exhibiting further palatalization/velarization in their forms. The surface forms that occur at stem-suffix boundaries are the result of ordered sound changes, as defined below.

1) VHR through glide formation.

2) Glide formation triggers a palatalization/velarization of the preceding consonant.

Evidence of such ordered sound changes has previously been documented by Krüger and Pretorius (2006) and Cole (1955). Palatalization and velarization are well-attested phonological processes in Setswana. Cole (1955) defines the following rules for palatalization and velarization, which accurately correspond with the 'varied' structures seen above.

Rules of Palatalization/Velarization (Cole, 1955)

1. If the consonant in the final syllable of the noun is bilabial (p, b, pʰ, f), it undergoes palatalization; if the final consonant is the bilabial nasal m, it undergoes velarization.
   b → **ɕʒʷ**; *kolobe* ('pig') → *kolojʷane* ('piglet')

   p → **tsʷ**; tshipi ('metal') → tshitsʷana ('small piece of iron')

   ph → **tʃʷ**; marophi ('blisters') → marotʃʷana ('small blisters')

   f → **tʃʷ**; gaufi ('nearby') → gautʃʷane ('fairly nearby')

   m → **ŋʷ**; molomo ('mouth') → moloŋʷana ('small mouth')

2. If the consonant in the final syllable of the noun is an alveolar (t, r, d, l, n) preceding a high front vowel, it undergoes palatalization; d also undergoes palatalization when followed by u.

t → **ts**; lobati (‘plank’) → lobatswana (‘small plank’)

r → **ts^{h}**; nare (‘buffalo’) → natshana (‘buffalo calf’)

du → **ts^{w}**; legodu (‘thief’) → legotswana (‘little thief’)

di→ **ts**; podi (‘goat’) → potsane (‘baby goat’)

n → **ɲ**; mosimane (‘boy’) → mosimaɲana (‘little boy’)

l → **j**; mmele (‘body’) → mmejana (‘little body’)

Palatalization and velarization in Setswana are a consequence of the incompatibility of bilabial, alveolar, velar, and glottal consonants with the glides w and j (Cole, 1955), and are found across many environments in the language. For instance, formation of passive verbs is achieved via suffixation of the morpheme *-wa* to the verb stem. This w in the suffix leads to palatalization/velarization of the preceding consonants, as shown below:

(35) a. *-lopa* (‘request’) + *-wa* → passive *-lotswa*

b. *-roba* (‘break’) + *-wa* → passive *-rojwa*

c. *-roma* (‘send’) + *-wa* → passive *-rongwa*

The glide -w is crucial in triggering such palatalization/velarization in the above passive forms. Suffixation in diminutives would not demonstrate similar palatalization/velarization if there were no ordered sound change. VHR through glide formation had to occur first, in order for the new glide to trigger a second-step palatalization/velarization of the preceding consonant. The question arises as to why glide formation in prefix-stem environments does not trigger the palatalization of preceding consonants. The answer might lie in whether VHR and palatalization/velarization are two different types of phonological alternations. Cole (1955) notes that these rules obtain except when the final vowel of the noun is an /-a/, resulting in a-a sequences.

(36) a. *t^{h}ìpànà*

t^{h}ìpà-ànà
9.knife-DIM
‘small knife’

b. *t^{h}àbànà*

t^{h}àbà-ànà
9.mountain-DIM
‘small mountain’

This paradigm seems to match the analysis of V1 = V2 sequences given in Section 4.5, wherein V1 elides in sequences of identical vowels.

5 Discussion

5.1 Why the Boundaries Differ

All three contexts are subject to the same trigger, *V.V, but they differ in what a repair is allowed to change. The pattern can be described as a scale of how strongly each vowel is protected.

At morpheme boundaries, V1 is the vowel of a prefix. Deleting or gliding a prefix vowel does not obscure the identity of the stem, so all repairs are available: V1 elision, glide formation, and diphthongization. Which repair applies depends on the features of the two vowels.

At syllable boundaries, both vowels belong to the same root. Root vowels appear to resist deletion: sequences such as /e̯/ + /i/ and /o̯/ + /u/, which elide at morpheme boundaries, are instead merged into a single syllable, which keeps both vowels. The only root-internal elision involves identical vowels, where deleting V1 loses no information. Glide formation, which also preserves the features of V1, is available in both word-internal contexts.

At word boundaries, both vowels are protected, and so is the syllable structure of each word. The two vowels belong to separate words, and syllabification does not appear to apply across the word boundary. Neither elision nor merging into one syllable is therefore possible, and the only available repair is to insert a consonant. Glottal stop epenthesis keeps both words intact, which also explains why both tones are retained.

Table 10 summarizes these conditions.

Table 10: **Conditions on VHR by Context**

Context	**Can V1 be deleted?**	**Can V1 and V2 share a syllable?**	**Repairs found**
Morpheme boundary (prefix-stem)	Yes	Yes	V1 elision, glide formation, diphthongization
Syllable boundary (within a root)	Only if identical to V2	Yes	Diphthongization, glide formation, elision of identical vowels
Word boundary	No	No	Glottal stop epenthesis

5.2 Gaps and Exceptions

The dataset covers most V1-V2 combinations (Tables 7 and 8), but some could not be tested. At morpheme boundaries, no examples were found for /i/ + /ɛ, ɔ, u/, /e̯/ + /o̯/, or /ɛ/ + /o̯/, and none of the prefixes examined end in /ɔ/ or /u/. At syllable boundaries, the attested combinations depend on the available vocabulary, so more

combinations remain untested.

A few patterns do not follow the general rules:

1. **/o̝/ + /ɔ/ elides at morpheme boundaries.** Since /ɔ/ is a homorganic lower vowel, the glide rule (Section 4.2) predicts glide formation, but the sequence surfaces as [ɔ]. One possibility is that the two vowels are similar enough to pattern with the V1 = V2 cases, given the overlap between the speaker's raised open-mid and close-mid vowels (Section 3.1). In citation forms, however, the two vowels are distinct (mean F1 of 426 vs. 593 Hz; Table 5), so this explanation is tentative. The same account may apply to /ɛ/ + /e̝/, which also elides.

2. **/ɛ/ + /u/ elides at morpheme boundaries,** whereas /e̝/ + /u/ undergoes glide formation. This may indicate that open-mid /ɛ/ is more prone to elision than close-mid /e̝/, consistent with its optional elision before /a/ and /ɔ/, but more examples are needed to confirm it.

3. **/ɛ/ + /a/ is optional in verbs, but only glide formation was recorded in nouns.** Since the optionality is free variation (Section 4.3), this difference may reflect the small number of noun items.

4. **/o̝/ + /i/ varies at syllable boundaries.** It diphthongizes in *búlòì* 'witchcraft' and *pòìtsʰèχò* 'phobia', but undergoes glide formation in *bòìdììdì* 'multitudes'.

5. **/ɛ/ + /ɔ/ diphthongizes at syllable boundaries** (*mòtʰèɔ* 'foundation'), although V2 is not higher than V1. Diphthongization at syllable boundaries may therefore also apply to vowels of equal height.

5.3 Limitations and Future Work

The data come from a single speaker of the Botswanan variety spoken in Phitshane Molopo. This allowed a large number of vowel combinations to be tested systematically, but some patterns may reflect individual or dialectal variation. For instance, the absence of glottal stop epenthesis at morpheme boundaries differs from the South African variety described by Bennett et al. (2016).

The classification of VHR strategies is based on auditory judgments, syllable tapping, humming of tone, and spectrographic inspection. Acoustic measurements (Section 4.7) support the main claims, but they cover the vowel inventory and a small number of tokens. Tone was likewise judged auditorily; the tonal generalization holds across all 147 word-internal items in the appendix (Tables 11–13), but the absence of contour tones on diphthongs remains a perceptual observation. In addition, many syllable-boundary items are loanwords, although the native words examined show the same patterns.

Future work could extend the analysis to more speakers and varieties, measure durations across more tokens at a controlled speech rate, compare F2 trajectories in diphthongs and glide + vowel sequences, and measure F0 on diphthongized syllables.

6 Conclusion

In conclusion, Setswana disallows vowel hiatus sequences and utilizes various VHR strategies to repair hiatus sequences that occur at different morpho-phonological boundaries. VHR in the language is triggered by a constraint that bans vowels in adjacent syllables (*V.V), while a ban on long vowels (*Vː) ensures that identical vowels surface as a single short vowel. Word-internal VHR is a regular-speed speech phenomenon, which when applied results in the re-syllabification of hetero-syllabic adjacent vowels into one CV syllable or separate CV.CV syllables. VHR in Setswana is conditioned by the features of the vowels involved, but more importantly by the context in which the hiatus sequence occurs. That is, while the features of the vowels determine the type of VHR – vowel elision, glide formation, diphthongization, or glottal stop epenthesis – it is the morpho-phonological context that further licenses or blocks the application of VHR. This results in variations in VHR across the different boundaries. The following is a brief summary of VHR in Setswana:

1. At morpheme boundaries, mid V1 vowels undergo elision when followed by a higher homorganic V2 vowel (e.g., /e̯/ + /i/, /o̯/ + /u/, /ɛ/ + /e̯/); /ɛ/ also elides before /u/.

2. High (/i and u/) or close-mid (/e̯ and o̯/) V1 vowels undergo glide formation when followed by any lower V2 vowel. Front V1 vowels become palatal glides [j], and back V1 vowels become labial glides [w].

3. Hiatus sequences involving a lower V1 vowel and a higher V2 vowel undergo diphthongization into one syllable at syllable boundaries. At morpheme boundaries, by contrast, this strategy is confined to sequences involving the low V1 vowel /a/ followed by any higher V2 vowel. Close-mid + high sequences, which elide at morpheme boundaries, become diphthongs at syllable boundaries.

4. In hiatus sequences of two identical vowels, V1 elides without compensatory lengthening, at both morpheme and syllable boundaries.

5. The front, open-mid vowel /ɛ/ exhibits an optionality in VHR, undergoing either glide formation or vowel elision when followed by /ɔ/ (in nouns and verbs) or /a/ (in verbs).

6. At word boundaries, vowel hiatus is always resolved using glottal stop epenthesis, regardless of the vowel combination.

Because Setswana is a tonal language, vowel hiatus produces CV-V structures with four possible tonal combinations, L-L, L-H, H-L, and H-H. VHR, thus, also impacts the tone on consecutive vowel segments. Since VHR causes the re-syllabification of hetero-syllabic vowels into mono-syllables, only one of the two tones surfaces after VHR. In brief, L-L tones surface as L; L-H and H-H tones surface as H; and H-L tones, as a result of HTS, surface as H. Interestingly, no contour tones were perceived on any of these sequences.

VHR at stem-suffix boundaries provides an interesting avenue for further research. Although palatalization and velarization are widely attested in the language, the palatal/velar forms that emerge seem rather unpredictable from the features of the consonants involved in the process. How and why these forms emerge, and whether they are driven by phonetic or lexical processes, requires further investigation. Additionally, vowel hiatus can occur between three consecutive vowels (Casali, 2011). This also needs to be studied and described.

References

Bab.la english-tswana dictionary. (n.d.). https://en.bab.la/dictionary/english-tswana/. (Accessed: 2024-25-04)

Baltazani, M. (2006). Focusing, prosodic phrasing, and hiatus resolution in greek. *Laboratory phonology*, *8*, 473–494.

Batibo, H. M. (2016). The origin and evolution of setswana culture: a linguistic account. *Botswana Notes and Records*, *48*, 134–149.

Bennett, W. G., Diemer, M., Kerford, J., Probert, T., & Wesi, T. (2016). Setswana (south african). *Journal of the International Phonetic Association*, *46*(2), 235–246.

Bickmore, L. (2021). Phonological and morphological influences on vowel hiatus resolution in rutooro. *Stellenbosch Papers in Linguistics Plus*, *62*(1), 39–55.

Casali, R. F. (1995). Patterns of glide formation in niger-congo: An optimality account.

Casali, R. F. (1996). A typology of vowel coalescence. *University of California Irvine Working Papers in Linguistics*, *2*, 29–42.

Casali, R. F. (1997). Vowel elision in hiatus contexts: Which vowel goes? *Language*, 493–533.

Casali, R. F. (2011). Hiatus resolution. *The Blackwell companion to phonology*, 1–27.

Casali, R. F. (2021). *Resolving hiatus*. Routledge.

Chebanne, A. M., Creissels, D., & Nkhwa, H. W. (1997). Tonal morphology of the setswana verb. *(No Title)*.

Cole, D. T. (1955). An introduction to tswana grammar.

Creissels, D. (1996). Reflex dataset 84:tswana.

De Haas, W. (2010). *A formal theory of vowel coalescence: a case study of ancient greek* (Vol. 30). Walter de Gruyter.

Faraclas, N. (1982). Elision and other morpheme boundary phenomena in the western dialects of obolo. *Journal of West African Languages*, *12*(2), 69–82.

Howard, I. (1972). *A directional theory of rule application in phonology.* (Unpublished doctoral dissertation). Massachusetts Institute of Technology.

Hyman, L. M. (2001). Privative tone in bantu. *Cross-linguistic studies of tonal phenomena*, 237–257.

Hyman, L. M. (2011). Tone: Is it different? *The handbook of phonological theory*, 197–239.

Inkelas, S. (2014). *The interplay of morphology and phonology* (Vol. 8). Oxford University Press.

Kadenge, M. (2010). Hiatus contexts and hiatus resolution strategies in zezuru. *Southern African Linguistics and Applied Language Studies*, *28*(1), 1–11.

Kadenge, M., & Simango, S. R. (2014). Comparing vowel hiatus resolution in cinsenga and chishona: An optimality theory analysis. *Stellenbosch Papers in Linguistics Plus*, *44*, 105–127.

Krüger, C. J. H., & Pretorius, R. S. (2006). *Introduction to the morphology of setswana* (Vol. 69). Lincom Europa.

Kutsch Lojenga, C. (1994). Ngiti: a central-sudanic language of zaire.

Lewis, M. P. (Ed.). (2009). *Ethnologue: Languages of the world* (Sixteenth ed.). Dallas, TX, USA: SIL International.

Lin, H.-s. (2018). Vowel hiatus resolution in kavalan. *Taiwan Journal of Linguistics*, *16*(1), 53–93.

Motzafi-Haller, P., Keuthmann, K., & Vossen, R. (2006). Setswana dialects and inter-dialectal variation in the republic of botswana. i. setswapong. *Zeitschrift der Deutschen Morgenländischen Gesellschaft*, *156*(1), 9–41.

Mudzingwa, C., & Kadenge, M. (2014). Coalescence as a hiatus resolution strategy in chikaranga–a dialect of chishona. *South African Journal of African Languages*, *34*(2), 127–136.

Overton, R. M. (2018). *Vowel hiatus resolution in simbiti and the orthographic implications* (Unpublished doctoral dissertation). MA thesis, Centre for Linguistics, Translation and Literacy: Radcliffe College.

Rialland, A., Aborobongui, M. E., Adda-Decker, M., & Lamel, L. (2015). Dropping of the class-prefix consonant, vowel elision and automatic phonological mining in embosi (bantu c 25). In *Selected proceedings of the 44th annual conference on african linguistics* (pp. 7–10).

Rosenthall, S. (1997). The distribution of prevocalic vowels. *Natural Language & Linguistic Theory*, *15*(1), 139–180.

Sabao, C. (2015). Resolving hiatus in (isi) ndebele: An optimality theoretic account. *Ampersand*, *2*, 122–135.

Sande, H. (2020). Morphologically conditioned phonology with two triggers. *Phonology*, *37*(3), 453–493.

Tsonope, J. (1988). *The acquisition of setswana noun class and agreement morphology–with special reference to demonstratives and possessives*. State University of New York at Buffalo.

Van de Velde, M., Bostoen, K., Nurse, D., & Philippson, G. (2019). The bantu languages.

Vratsanos, A., & Kadenge, M. (2017). Hiatus resolution in xitsonga. *Stellenbosch Papers in Linguistics Plus*, *52*, 175–196.

Yip, M. J. W. (2002). *Tone*. Cambridge University Press.

Appendix

A VHR at Morpheme Boundaries

Table 11: **VHR at Prefix-Stem Boundaries (SM-Verb Stem)**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
1.	a-a	bá-àpàjà nàmà	bápájá nàmà	They cook meat.	V1 Elision	H-L	H (HTS)
2.	a-a	bá-ádìmà bùkà	bádímá bùkà	They lend a book.	V1 Elision	H-H	H (HTS)
3.	a-ɛ	bá-ɛ̀tà xàè	báɛtá xàè	They visit home.	Diphthongization	H-L	H (HTS)
4.	a-ɛ	bá-ɛ̀pà m̀mù	báɛpá m̀mù	They dig soil.	Diphthongization	H-L	H (HTS)
5.	a-e	bá-émà fà	báemá fà	They stand here.	Diphthongization	H-H	H (HTS)
6.	a-ɔ	bá-ɔ̀pèlà dì-pìnà	báɔpélá dì-pìnà	They sing songs.	Diphthongization	H-L	H (HTS)
7.	a-ɔ	bá-ɔ̀lèlà m̀mù	báɔlélá m̀mù	They pick up dirt.	Diphthongization	H-L	H (HTS)
8.	a-i	bá-íkàŋà ròná	báikáŋá ròná	They depend on us.	Diphthongization	H-H	H (HTS)
9.	a-u	bá-útɬwà módùmò	báutɬwá mód̀ùmò	They hear noise.	Diphthongization	H-H	H (HTS)
10.	e-a	kè-àpàjà nàmà	kʲàpàjà nàmà	I cook meat.	Glide Formation	L-L	L
11.	e-a	kè-àlà bòlàò	kʲàlà bòlàò	I make the bed.	Glide Formation	L-L	L
12.	e-ɛ	kè-ɛ̀tà bùsìχ̀ò	kʲɛ̀tà bùsìχ̀ò	I visit at night.	Glide Formation	L-L	L
13.	e-ɛ	kè-ɛ̀pà m̀mù	kʲɛ̀pà m̀mù	I dig soil.	Glide Formation	L-L	L
14.	e-e	kè-émà fà	kémà fà	I stand here.	V1 Elision	L-H	H

Table 11: **VHR at Prefix-Stem Boundaries (SM-Verb Stem)**

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
15.	e-ɔ	kè-ɔ̀pèlà dì-pìnà	kʲɔ̀pèlà dì-pìnà	I sing songs.	Glide Formation	L-L	L
16.	e-ɔ	kè-ɔ̀lèlà m̀mù	kʲɔ̀lèlà m̀mù	I pick up dirt.	Glide Formation	L-L	L
17.	e-i	kè-íkàɲà bònè	kíkàɲà bònè	I depend on them.	V1 Elision	L-H	H
18.	e-u	kè-útɬwà módùmò	kʲútɬwà mó́dùmò	I hear noise.	Glide Formation	L-H	H
19.	ɛ-a ɛ-a	rɛ̀-àpàjà nàmà rɛ̀-àpàjà nàmà	rʲàpàjà nàmà ràpàjà nàmà	We cook meat.	Glide Formation V1 Elision	L-L L-L	L L
20.	ɛ-ɛ	rɛ̀-ɛ̀tà bùsìχò	rɛ̀tà bùsìχò	We visit at night.	V1 Elision	L-L	L
21.	ɛ-ɛ	rɛ̀-ɛ̀pà m̀mù	rɛ̀pà m̀mù	We dig soil.	V1 Elision	L-L	L
22.	ɛ-e	rɛ̀-émà fà	rémà fà	We stand here.	V1 Elision	L-H	H
23.	ɛ-ɔ ɛ-ɔ	rɛ̀-ɔ̀pèlà dì-pìnà rɛ̀-ɔ̀pèlà dì-pìnà	rʲɔ̀pèlà dì-pìnà rɔ̀pèlà dì-pìnà	We sing songs.	Glide Formation V1 Elision	L-L L-L	L L
24.	ɛ-ɔ ɛ-ɔ	rɛ̀-ɔ̀lèlà m̀mù rɛ̀-ɔ̀lèlà m̀mù	rʲɔ̀lèlà m̀mù rɔ̀lèlà m̀mù	We pick up dirt.	Glide Formation V1 Elision	L-L L-L	L L
25.	ɛ-i	rɛ̀-íkàɲà bònè	ríkàɲà bònè	We depend on them.	V1 Elision	L-H	H
26.	ɛ-u	rɛ̀-útɬwà módùmò	rútɬwà mó́dùmò	We hear noise.	V1 Elision	L-H	H
27.	o-a	χò-àpàjà	χʷàpàjà	To cook.	Glide Formation	L-L	L
28.	o-a	χò-àlà	χʷàlà	To make bed.	Glide Formation	L-L	L
29.	o-a	χò-ádìmà	χʷádìmà	To lend/borrow.	Glide Formation	L-H	H
30.	o-ɛ	χò-ɛ̀tà	χʷɛ̀tà	To visit.	Glide Formation	L-L	L
31.	o-ɛ	χò-ɛ̀pà	χʷɛ̀pà	To dig.	Glide Formation	L-L	L

Table 11: **VHR at Prefix-Stem Boundaries (SM-Verb Stem)**

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
32.	o-e	χò-émà	χ^{w}émà	To stand.	Glide Formation	L-H	H
33.	o-ɔ	χò-ɔ̀pèlà	χɔ̀pèlà	To sing.	V1 Elision	L-L	L
34.	o-ɔ	χò-ɔ̀lèlà	χɔ̀lèlà	To pick up.	V1 Elision	L-L	L
35.	o-i	χò-íkàŋà	χ^{w}íkàŋà	To depend on.	Glide Formation	L-H	H
36.	o-u	χò-útɬwà	χútɬwà	To hear.	V1 Elision	L-H	H
37.	o-a	ò-àpàjà nàmà	wàpàjà nàmà	You cook meat.	Glide Formation	L-L	L
38.	o-a	ò-ádìmà bùkà	wádìmà bùkà	You lend a book.	Glide Formation	L-H	H
39.	o-ɛ	ò-ɛ̀tà bùsìxò	wɛ̀tà bùsìxò	You visit at night.	Glide Formation	L-L	L
40.	o-ɛ	ò-ɛ̀pà m̀mù	wɛ̀pà m̀mù	You dig soil.	Glide Formation	L-L	L
41.	o-e	ò-émà fà	wémà fà	You stand here.	Glide Formation	L-H	H
42.	o-ɔ	ò-ɔ̀pèlà dì-pìnà	ɔ̀pèlà dì-pìnà	You sing songs.	V1 Elision	L-L	L
43.	o-ɔ	ò-ɔ̀lèlà m̀mù	ɔ̀lèlà m̀mù	You pick up dirt.	V1 Elision	L-L	L
44.	o-i	ò-íkàŋà bònè	wíkàŋà bònè	You depend on them.	Glide Formation	L-H	H
45.	o-u	ò-útɬwà módùmò	útɬwà mó́dùmò	You hear noise.	V1 Elision	L-H	H
46.	o-a	ó-àpàjà nàmà	wápájá nàmà	S/he cooks meat.	Glide Formation	H-L	H (HTS)
47.	o-a	ó-ádìmà bùkà	wádímá bùkà	S/he lends a book.	Glide Formation	H-H	H (HTS)
48.	o-ɛ	ó-ɛ̀tà bùsìχò	wɛ́tá bùsìχò	S/he visits at night.	Glide Formation	H-L	H (HTS)

Table 11: **VHR at Prefix-Stem Boundaries (SM-Verb Stem)**

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
49.	o-ɛ	ó-ɛ̀pà m̀mù	wɛ́pá m̀mù	S/he digs soil.	Glide Formation	H-L	H (HTS)
50.	o-e	ó-émà fà	wémà fà	S/he stands here.	Glide Formation	H-H	H (HTS)
51.	o-ɔ	ó-ɔ́pèlà dì-pìnà	ɔ́pélá dì-pìnà	S/he sings songs.	V1 Elision	H-L	H (HTS)
52.	o-ɔ	ó-ɔ̀lèlà m̀mù	ɔ́lélá m̀mù	S/he picks up dirt.	V1 Elision	H-L	H (HTS)
53.	o-i	ó-íkàɲà bònè	wíkáɲá bònè	S/he depends on them.	Glide Formation	H-H	H (HTS)
54.	o-u	ó-útɬwà módùmò	útɬwá módùmò	S/he hears noise.	V1 Elision	H-H	H (HTS)

Table 12: **VHR at Prefix-Stem Boundaries (NC-Noun Stem)**

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
1.	a-a	mà-ànò	mànò	Plans	V1 Elision	L-L	L
2.	a-a	mà-àróχì	màróχì	Rebels	V1 Elision	L-L	L
3.	a-a	bá-áχì	báχì	Builders	V1 Elision	H-H	H
4.	a-a	bá-áχèdì	báχèdì	Citizens	V1 Elision	H-H	H
5.	a-ɛ	mà-ɛ̀tò	màɛtò	Journeys	Diphthongization	L-L	L
6.	a-ɛ	mà-ɛ̀bà	màɛbà	Doves	Diphthongization	L-L	L
7.	a-e	bá-ètsì	báetsì	Imitators	Diphthongization	H-L	H
8.	a-e	bá-èǹg	báeǹg	Visitors	Diphthongization	H-L	H
9.	a-e	bá-èmìsì	báemìsì	Bridesmaids	Diphthongization	H-L	H

Table 12: **VHR at Prefix-Stem Boundaries (NC-Noun Stem)**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
10.	a-ɔ	mà-ɔ̀bò	màɔbò	Boundaries	Diphthongization	L-L	L
11.	a-ɔ	mà-ɔ́tò	máɔtò	Legs	Diphthongization	L-H	H
12.	a-o	bá-òkì	báokì	Nurses	Diphthongization	H-L	H
13.	a-o	bá-òkàmédì	báokàmédì	Supervisors	Diphthongization	H-L	H
14.	a-i	mà-ísò	máisò	Fireplaces	Diphthongization	L-H	H
15.	a-i	mà-íkàtɬò	máikàtɬò	Regrets	Diphthongization	L-H	H
16.	a-i	bá-ìtùdì	báitùdì	Students	Diphthongization	H-L	H
17.	a-u	mà-úbà	máubà	Droughts	Diphthongization	L-H	H
18.	ɛ-a	sɛ̀-ànè	sʲànè	Totem	Glide Formation	L-L	L
19.	ɛ-a	sɛ̀-àtɬà	sʲàtɬà	Hand	Glide Formation	L-L	L
20.	ɛ-a	sɛ̀-àpàrò	sʲàpàrò	Clothing	Glide Formation	L-L	L
21.	ɛ-a	lɛ̀-ànò	lʲànò	Plan	Glide Formation	L-L	L
22.	ɛ-a	lɛ̀-àróχì	lʲàróχì	Rebel	Glide Formation	L-H	H
23.	ɛ-ɛ	sɛ̀-ɛ̀lè	sɛ̀lè	Sentence	V1 Elision	L-L	L
24.	ɛ-ɛ	sɛ̀-ɛ̀mò	sɛ̀mò	Situation	V1 Elision	L-L	L
25.	ɛ-ɛ	lɛ̀-ɛ̀tò	lɛ̀tò	Journey	V1 Elision	L-L	L
26.	ɛ-ɛ	lɛ̀-ɛ̀bà	lɛ̀bà	Dove	V1 Elision	L-L	L
27.	ɛ-ɔ	sɛ̀-ɔ́pò	sʲɔ́pò		Glide Formation	L-H	H
		sɛ̀-ɔ́pò	sɔ́pò	Group	V1 Elision	L-H	H
28.	ɛ-ɔ	sɛ̀-ɔ̀pèdì	sʲɔ̀pèdì		Glide Formation	L-L	L
		sɛ̀-ɔ̀pèdì	sɔ̀pèdì	Singer	V1 Elision	L-L	L
29.	ɛ-ɔ	lɛ̀-ɔ̀bò	lʲɔ̀bò		Glide Formation	L-L	L
		lɛ̀-ɔ́bò	lɔ́bò	Boundary	V1 Elision	L-H	H
30.	ɛ-ɔ	lɛ̀-ɔ́tò	lʲɔ́tò		Glide Formation	L-H	H
		lɛ̀-ɔ́tò	lɔ́tò	Leg	V1 Elision	L-H	H
31.	ɛ-i	sɛ̀-ìpátì	sìpátì	Excuse	V1 Elision	L-L	L

Table 12: **VHR at Prefix-Stem Boundaries (NC-Noun Stem)**

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
32.	ɛ-i	sɛ̀-ìpònè	sìpònè	Mirror	V1 Elision	L-L	L
33.	ɛ-i	lɛ̀-ísò	lísò	Fireplace	V1 Elision	L-H	H
34.	ɛ-i	lɛ̀-ìtɬò	lìtɬò	Eye	V1 Elision	L-L	L
35.	ɛ-u	lɛ̀-úbà	lúbà	Drought	V1 Elision	L-H	H
36.	e-a	mè-àlò	mʲàlò	Coverings	Glide Formation	L-L	L
37.	e-a	mè-àχò	mʲàχò	Buildings	Glide Formation	L-L	L
38.	e-e	mè-èpò	mèpò	Mines	V1 Elision	L-L	L
39.	e-i	mè-ìlà	mìlà	Taboos	V1 Elision	L-L	L
40.	e-u	mè-úwànè	mʲúwànè	Mists	Glide Formation	L-H	H
41.	o-a	mò-àlò	mʷàlò	Covering	Glide Formation	L-L	L
42.	o-a	mò-áxì	mʷáxì	Builder	Glide Formation	L-H	H
43.	o-a	mò-áχèdì	mʷáχèdì	Citizen	Glide Formation	L-H	H
44.	o-e	mò-èmìsì	mʷèmìsì	Bridesmaid	Glide Formation	L-L	L
45.	o-e	mò-èpò	mʷèpò	Mine	Glide Formation	L-L	L
46.	o-e	mò-èǹg	mʷèǹg	Visitor	Glide Formation	L-L	L
47.	o-e	mò-ètsì	mʷètsì	Imitator	Glide Formation	L-L	L
48.	o-o	mò-òkì	mòkì	Nurse	V1 Elision	L-L	L
49.	o-o	mò-òkàmédì	mòkàmédì	Supervisor	V1 Elision	L-L	L
50.	o-i	mò-ìlà	mʷìlà	Taboo	Glide Formation	L-L	L
51.	o-i	mò-ìtùdì	mʷìtùdì	Student	Glide Formation	L-L	L
52.	o-u	mò-úwànè	múwànè	Mist	V1 Elision	L-H	H
53.	i-a	dì-ànè	dʲànè	Totems	Glide Formation	L-L	L
54.	i-a	dì-àpàrò	dʲàpàrò	Clothes	Glide Formation	L-L	L
55.	i-a	dì-àtɬà	dʲàtɬà	Hands	Glide Formation	L-L	L
56.	i-e	dì-èlè	dʲèlè	Sentences	Glide Formation	L-L	L
57.	i-e	dì-èmò	dʲèmò	Situations	Glide Formation	L-L	L

Table 12: **VHR at Prefix-Stem Boundaries (NC-Noun Stem)**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
58.	i-o	dì-òmìsì	dʲòmìsì	Driers	Glide Formation	L-L	L
59.	i-i	dì-ìpátì	dìpátì	Excuses	V1 Elision	L-L	L
60.	i-i	dì-ìpònè	dìpònè	Mirrors	V1 Elision	L-L	L

B VHR at Syllable Boundaries

Table 13: **VHR at Syllable Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
1.	a-a	sà.à.tà.nì	sà.tà.nì	Satan	V1 Elision	L-L	L
2.	a-ɛ	mú.là.ɛ̀.tsà	múlàɛtsà	Message	Diphthongization	L-L	L
3.	a-e	m̀mú.sà.è.sì	m̀mú.sàe.sì	Dictator	Diphthongization	L-L	L
4.	a-o	mò.là.ò	mò.lào	Law	Diphthongization	L-L	L
5.	a-o	là.ò.lè.là	lào.lè.là	Allow	Diphthongization	L-L	L
6.	a-i	m̀mà.ì.lè	m̀mài.lè	Mile	Diphthongization	L-L	L
7.	a-u	sè.mà.ù.sù	sè.màu.sù	Kiosk	Diphthongization	L-L	L
8.	a-u	sì.kà.ù.sù	sì.kàu.sù	Socks	Diphthongization	L-L	L
9.	a-u	gà.ú.tà	gáu.tà	Gold	Diphthongization	L-H	H
10.	ɛ-ɛ	nɛ̀.ɛ̀.lò	nɛ̀.lò	Offer	V1 Elision	L-L	L
11.	ɛ-ɔ	mò.tʰɛ̀. ɔ̀	mò.tʰɛ̀ɔ	Foundation	Diphthongization	L-L	L
12.	e-ɔ	pè.ɔ̀.mù.là.tò	pʲɔ̀.mù.là.tò	Accusation	Glide Formation	L-L	L
13.	e-ɔ	tsè.ɔ̀.kà.rù.lò	tsʲɔ̀.kà.rù.lò	Participation	Glide Formation	L-L	L
14.	e-i	tʰè.í.pì	tʰéi.pì	Tape	Diphthongization	L-H	H
15.	e-u	bà.bè.ù.dì	bà.bʲù.dì	Barber	Glide Formation	L-L	L
16.	o-a	bò.bò.à.ɲà.nà	bò.bʷà.ɲà.nà	Fluffy	Glide Formation	L-L	L

Table 13: **VHR at Syllable Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
17.	o-e	pò.è.lètsò	p^{w}è.lè.tsò	Repetition	Glide Formation	L-L	L
18.	o-i	bú.lò.ì	bú.lòi	Witchcraft	Diphthongization	L-L	L
19.	o-i	pò.ì.ts^{h}è.χò	pòi.ts^{h}è.χò	Phobia	Diphthongization	L-L	L
20.	o-u	só.ù.sò	só.ù.sò	Sauce	Diphthongization	H-L	H
21.	o-u	pò.ù.nà.mà	pòu.nà.mà	Lip	Diphthongization	L-L	L
22.	i-a	mì.ú.sì.á.mù	m^{j}ú.s^{j}á.mù	Museum	Glide Formation	L-H	H
23.	i-e	rà.dì.e.tà.rá	rá.d^{j}è.tà.rà	Radiator	Glide Formation	L-L	L
24.	i-o	dì.fì.o.sè	dì.f^{j}ò.sè	Fuse	Glide Formation	L-L	L
25.	i-i	tì.ì	tì	Tea	V1 Elision	L-L	L
26.	o-i	bò.ì.dì.ì.dì	b^{w}ì.dì.dì	Multitudes	Glide Formation	L-L	L
27.	i-i	bò.ì.dì.ì.dì	b^{w}ì.dì.dì	Multitudes	V1 Elision	L-L	L
28.	i-u	k^{h}ò.m.pì.ù.tà.rà	k^{h}ò.m.p^{j}ù.tà.rà	Computer	Glide Formation	L-L	L
29.	u-a	kù.kù.à.nà	kù.k^{w}à.nà	Gathering	Glide Formation	L-L	L
30.	u-e	bù.ì.pù.é.lè.tsò	b^{w}ì.p^{w}é.lè.tsò	Repetition	Glide Formation	L-H	H
31.	u-i	bù.ì.pù.é.lè.tsò	b^{w}ì.p^{w}é.lè.tsò	Repetition	Glide Formation	L-L	L
32.	u-i	pù.ì.sà.ɲó	p^{w}ì.sà.ɲó	Dialogue	Glide Formation	L-L	L
33.	u-u	sè.kù.ù.pù	sè.kù.pù	Bottle	V1 Elision	L-L	L

C VHR at Word Boundaries

Table 14: **VHR at Word Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
1.	a-a	kè-bɔ̀nà àpòlè	kè-bɔ̀nà^{ʔ}àpòlè	I see an apple.	Epenthesis - Glottal Stop	L-L	L-L

Table 14: **VHR at Word Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
2.	a-a	kè-bɔ̀nà àɲànsì	kè-bɔ̀nàʔàɲànsì	I see an onion.	Epenthesis - Glottal Stop	L-L	L-L
3.	a-ɛ	kè-bɔ̀nà ɛ̀mɛ̀rɛ̀	kè-bɔ̀nàʔɛ̀mɛ̀rɛ̀	I see a bucket.	Epenthesis - Glottal Stop	L-L	L-L
4.	a-ɛ	kè-bɔ̀nà ɛ̀nʤɛ̀nɛ̀	kè-bɔ̀nàʔɛ̀nʤɛ̀nɛ̀	I see an engine.	Epenthesis - Glottal Stop	L-L	L-L
5.	a-o	kè-bɔ̀nà ɔ̀fîsì	kè-bɔ̀nàʔɔ̀fîsì	I see an office.	Epenthesis - Glottal Stop	L-L	L-L
6.	a-ɔ	kè-bɔ̀nà ɔ̀vɛ̀nɛ̀	kè-bɔ̀nàʔɔ̀vɛ̀nɛ̀	I see an oven.	Epenthesis - Glottal Stop	L-L	L-L
7.	a-ɔ	kè-bɔ̀nà ɔ̀mfɔ̀lɔ̀pɔ	kè-bɔ̀nàʔɔ̀mfɔ̀lɔ̀pɔ̀	I see an envelope.	Epenthesis - Glottal Stop	L-L	L-L
8.	a-i	kè-bɔ̀nà ísàχò	kè-bɔ̀nàʔísàχò	I see Isago.	Epenthesis - Glottal Stop	L-H	L-H
9.	a-u	kè-bɔ̀nà úbànì	kè-bɔ̀nàʔúbànì	I see Ubani.	Epenthesis - Glottal Stop	L-L	L-L
10.	ɛ-a	kè-rɛ̀kìlɛ̀ àpòlè	kè-rɛ̀kìlɛ̀ʔàpòlè	I bought an apple.	Epenthesis - Glottal Stop	L-L	L-L
11.	ɛ-a	kè-rɛ̀kìlɛ̀ àɲànsì	kè-rɛ̀kìlɛ̀ʔàɲànsì	I bought an onion.	Epenthesis - Glottal Stop	L-L	L-L
12.	ɛ-ɛ	kè-rɛ̀kìlɛ̀ ɛ̀mɛ̀rɛ̀	kè-rɛ̀kìlɛ̀ʔɛ̀mɛ̀rɛ̀	I bought a bucket.	Epenthesis - Glottal Stop	L-L	L-L
13.	ɛ-ɛ	kè-rɛ̀kìlɛ̀ ɛ̀nʤɛ̀nɛ̀	kè-rɛ̀kìlɛ̀ʔɛ̀nʤɛ̀nɛ̀	I bought an engine.	Epenthesis - Glottal Stop	L-L	L-L
14.	ɛ-o	kè-rɛ̀kìlɛ̀ ɔ̀fîsì	kè-rɛ̀kìlɛ̀ʔɔ̀fîsì	I bought an office.	Epenthesis - Glottal Stop	L-L	L-L

Table 14: **VHR at Word Boundaries**

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
15.	ɛ-ɔ	kè-rɛ̀kìlɛ̀ ɔ̀vɛ̀nɛ̀	kè-rɛ̀kìlɛ̀ʔɔ̀vɛ̀nɛ̀	I bought an oven.	Epenthesis - Glottal Stop	L-L	L-L
16.	ɛ-ɔ	kè-rɛ̀kìlɛ̀ ɔ̀mfɔ̀lɔ̀pɔ	kè-rɛ̀kìlɛ̀ʔɔ̀mfɔ̀lɔ̀pɔ	I bought an envelope.	Epenthesis - Glottal Stop	L-L	L-L
17.	ɛ-i	kè-rɛ̀kɛ̀tsɛ̀ ísàχò pènè	kè-rɛ̀kɛ̀tsɛ̀ʔísàχò pènè	I bought Isago a pen.	Epenthesis - Glottal Stop	L-H	L-H
18.	ɛ-u	kè-rɛ̀kɛ̀tsɛ̀ úbànì pènè	kè-rɛ̀kɛ̀tsɛ̀ʔúbànì pènè	I bought Ubani a pen.	Epenthesis - Glottal Stop	L-L	L-L
19.	e-a	kè-fà mòsìmànè àpòlè	kè-fà mòsìmànèʔàpòlè	I give the boy an apple.	Epenthesis - Glottal Stop	L-L	L-L
20.	e-a	kè-fà mòsìmànè àɲànsì	kè-fà mòsìmànèʔàɲànsì	I give the boy an onion.	Epenthesis - Glottal Stop	L-L	L-L
21.	e-ɛ	kè-fà mòsìmànè ɛ̀mɛ̀rɛ̀	kè-fà mòsìmànèʔɛ̀mɛ̀rɛ̀	I give the boy a bucket.	Epenthesis - Glottal Stop	L-L	L-L
22.	e-ɛ	kè-fà mòsìmànè ɛ̀nʤɛ̀nɛ̀	kè-fà mòsìmànèʔɛ̀nʤɛ̀nɛ̀	I give the boy an engine.	Epenthesis - Glottal Stop	L-L	L-L
23.	e-o	kè-fà mòsìmànè òfìsì	kè-fà mòsìmànèʔòfìsì	I give the boy an office.	Epenthesis - Glottal Stop	L-L	L-L
24.	e-ɔ	kè-fà mòsìmànè ɔ̀vɛ̀nɛ̀	kè-fà mòsìmànèʔɔ̀vɛ̀nɛ̀	I give the boy an oven.	Epenthesis - Glottal Stop	L-L	L-L
25.	e-ɔ	kè-fà mòsìmànè ɔ̀mfɔ̀lɔ̀pɔ	kè-fà mòsìmànèʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I give the boy an envelope.	Epenthesis - Glottal Stop	L-L	L-L
26.	e-i	kè-fà mòsìmànè ísàχò	kè-fà mòsìmànèʔísàχò	I give the boy Isago.	Epenthesis - Glottal Stop	L-H	L-H
27.	e-u	kè-fà mòsìmànè úbànì	kè-fà mòsìmànèʔúbànì	I give the boy Ubani.	Epenthesis - Glottal Stop	L-L	L-L

Table 14: **VHR at Word Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
28.	e-ɛ	rè-ádìmà mòsìmànè èmèrè	rádìmà mòsìmànèʔèmèrè	We lend the boy a bucket.	Epenthesis - Glottal Stop	L-L	L-L
29.	e-ɔ	ó-ádìmà mòsìmànè ɔ̀vènè	wádímá mòsìmànèʔɔ̀vènè	S/he lends the boy an oven.	Epenthesis - Glottal Stop	L-L	L-L
30.	ɔ-a	kè-fà tʰèɔ̀ àpòlè	kè-fà tʰèɔ̀ʔàpòlè	I give Theo an apple.	Epenthesis - Glottal Stop	L-L	L-L
31.	ɔ-a	kè-fà tʰèɔ̀ àɲànsì	kè-fà tʰèɔ̀ʔàɲànsì	I give Theo an onion.	Epenthesis - Glottal Stop	L-L	L-L
32.	ɔ-ɛ	kè-fà tʰèɔ̀ èmèrè	kè-fà tʰèɔ̀ʔèmèrè	I give Theo a bucket.	Epenthesis - Glottal Stop	L-L	L-L
33.	ɔ-ɛ	kè-fà tʰèɔ̀ ènɠènè	kè-fà tʰèɔ̀ʔènɠènè	I give Theo an engine.	Epenthesis - Glottal Stop	L-L	L-L
34.	ɔ-o	kè-fà tʰèɔ̀ òfìsì	kè-fà tʰèɔ̀ʔòfìsì	I give Theo an office.	Epenthesis - Glottal Stop	L-L	L-L
35.	ɔ-ɔ	kè-fà tʰèɔ̀ɔ̀vènè	kè-fà tʰèɔ̀ʔɔ̀vènè	I give Theo an oven.	Epenthesis - Glottal Stop	L-L	L-L
36.	ɔ-ɔ	kè-fà tʰèɔ̀ ɔ̀mfɔ̀lɔ̀pɔ	kè-fà tʰèɔ̀ʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I give Theo an envelope.	Epenthesis - Glottal Stop	L-L	L-L
37.	ɔ-i	kè-fà tʰèɔ̀ ísàχò	kè-fà tʰèɔ̀ʔísàχò	I give Theo Isago.	Epenthesis - Glottal Stop	L-H	L-H
38.	ɔ-u	kè-fà tʰèɔ̀ úbànì	kè-fà tʰèɔ̀ʔúbànì	I give Theo Ubani.	Epenthesis - Glottal Stop	L-L	L-L
39.	ɔ-ɛ	rè-ádìmà tʰèɔ̀ èmèrè	ràdìmà tʰèɔ̀ʔèmèrè	We lend Theo a bucket.	Epenthesis - Glottal Stop	L-L	L-L

Table 14: **VHR at Word Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
40.	ɔ-ɔ	ó-ádìmà t^{h}èɔ̀ ɔ̀vɛ̀nɛ̀	wádímá t^{h}èɔ̀ʔɔ̀vɛ̀nɛ̀	S/he lends Theo an oven.	Epenthesis - Glottal Stop	H-L	H-L
41.	o-a	kè-fà ísàxò àpòlè	kè-fà ísàxòʔàpòlè	I give Isago an apple.	Epenthesis - Glottal Stop	L-L	L-L
42.	o-a	kè-fà ísàχò àɲànsì	kè-fà ísàχòʔàɲànsì	I give Isago an onion.	Epenthesis - Glottal Stop	L-L	L-L
43.	o-ɛ	kè-fà ísàχò èmèrɛ̀	kè-fà ísàχòʔèmèrɛ̀	I give Isago a bucket.	Epenthesis - Glottal Stop	L-L	L-L
44.	o-ɛ	kè-fà ísàχò ènʤènɛ̀	kè-fà ísàχòʔènʤènɛ̀	I give Isago an engine.	Epenthesis - Glottal Stop	L-L	L-L
45.	o-o	kè-fà ísàχò òfìsì	kè-fà ísàχòʔòfìsì	I give Isago an office.	Epenthesis - Glottal Stop	L-L	L-L
46.	o-ɔ	kè-fà ísàχò ɔ̀vɛ̀nɛ̀	kè-fà ísàχòʔɔ̀vɛ̀nɛ̀	I give Isago an oven.	Epenthesis - Glottal Stop	L-L	L-L
47.	o-ɔ	kè-fà ísàχò ɔ̀mfɔ̀lɔ̀pɔ	kè-fà ísàχòʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I give Isago an envelope.	Epenthesis - Glottal Stop	L-L	L-L
48.	o-i	kè-fà ísàχò ísàχɔ̀	kè-fà ísàχòʔísàχɔ̀	I give Isago Isago.	Epenthesis - Glottal Stop	L-H	L-H
49.	o-u	kè-fà ísàχò úbànì	kè-fà ísàχòʔúbànì	I give Isago Ubani.	Epenthesis - Glottal Stop	L-L	L-L
50.	o-ɛ	rè-ádìmà ísàχò èmèrɛ̀	ràdìmà ísàχòʔèmèrɛ̀	We lend Isago a bucket.	Epenthesis - Glottal Stop	L-L	L-L
51.	o-ɔ	ó-ádìmà ísàχò ɔ̀vɛ̀nɛ̀	wádímá ísàχòʔɔ̀vɛ̀nɛ̀	S/he lends Isago an oven.	Epenthesis - Glottal Stop	H-L	H-L
52.	i-a	kè-fìlɛ̀ mòsàdì àpòlè	kè-fìlɛ̀ mòsàdìʔàpòlè	I gave the woman an apple.	Epenthesis - Glottal Stop	L-L	L-L

Table 14: **VHR at Word Boundaries**

S.no	V1-V2	Underlying Form	Surface Form	Translation	VHR Strategy	V1-V2 Tone	Surface Tone
53.	i-a	kè-fílè mòsàdì àɲànsì	kè-fílè mòsàdìʔàɲànsì	I gave the woman an onion.	Epenthesis - Glottal Stop	L-L	L-L
54.	i-ɛ	kè-fílè mòsàdì èmɛ̀rɛ̀	kè-fílè mòsàdìʔèmɛ̀rɛ̀	I gave the woman a bucket.	Epenthesis - Glottal Stop	L-L	L-L
55.	i-ɛ	kè-fílè mòsàdì ènʤènè	kè-fílè mòsàdìʔènʤènè	I gave the woman an engine.	Epenthesis - Glottal Stop	L-L	L-L
56.	i-o	kè-fílè mòsàdì òfìsì	kè-fílè mòsàdìʔòfìsì	I gave the woman an office.	Epenthesis - Glottal Stop	L-L	L-L
57.	i-ɔ	kè-fílè mòsàdì ɔ̀vɛ̀nɛ̀	kè-fílèmòsàdìʔɔ̀vɛ̀nɛ̀	I gave the woman an oven.	Epenthesis - Glottal Stop	L-L	L-L
58.	i-ɔ	kè-fílè mòsàdì ɔ̀mfɔ̀lɔ̀pɔ	kè-fílè mòsàdìʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I gave the woman an envelope.	Epenthesis - Glottal Stop	L-L	L-L
59.	i-i	kè-fílè mòsàdì ísàχɔ̀	kè-fílè mòsàdìʔísàχɔ̀	I gave the woman Isago.	Epenthesis - Glottal Stop	L-H	L-H
60.	i-u	kè-fílè mòsàdì úbànì	kè-fílèmòsàdìʔúbànì	I gave the woman Ubani.	Epenthesis - Glottal Stop	L-L	L-L
61.	u-a	kè-fílè mòtʰù àpòlè	kè-fílè mòtʰùʔàpòlè	I gave the person an apple.	Epenthesis - Glottal Stop	L-L	L-L
62.	u-a	kè-fílè mòtʰù àɲànsì	kè-fílè mòtʰùʔàɲànsì	I gave the person an onion.	Epenthesis - Glottal Stop	L-L	L-L
63.	u-ɛ	kè-fílè mòtʰù èmɛ̀rɛ̀	kè-fílè mòtʰùʔèmɛ̀rɛ̀	I gave the person a bucket.	Epenthesis - Glottal Stop	L-L	L-L
64.	u-ɛ	kè-fílè mòtʰù ènʤènè	kè-fílè mòtʰùʔènʤènè	I gave the person an engine.	Epenthesis - Glottal Stop	L-L	L-L
65.	u-o	kè-fílè mòtʰù òfìsì	kè-fílè mòtʰùʔòfìsì	I gave the person an office.	Epenthesis - Glottal Stop	L-L	L-L

Table 14: **VHR at Word Boundaries**

S.no	**V1-V2**	**Underlying Form**	**Surface Form**	**Translation**	**VHR Strategy**	**V1-V2 Tone**	**Surface Tone**
66.	u-ɔ	kè-fìlɛ̀ mòtʰù ɔ̀vɛ̀nɛ̀	kè-fìlɛ̀mòtʰùʔɔ̀vɛ̀nɛ̀	I gave the person an oven.	Epenthesis - Glottal Stop	L-L	L-L
67.	u-ɔ	kè-fìlɛ̀ mòtʰù ɔ̀mfɔ̀lɔ̀pɔ	kè-fìlɛ̀ mòtʰùʔ ɔ̀mfɔ̀lɔ̀pɔ̀	I gave the person an envelope.	Epenthesis - Glottal Stop	L-L	L-L
68.	u-i	kè-fìlɛ̀ mòtʰù ísàχɔ̀	kè-fìlɛ̀ mòtʰùʔísàχɔ̀	I gave the person Isago.	Epenthesis - Glottal Stop	L-H	L-H
69.	u-u	kè-fìlɛ̀ mòtʰù úbànì	kè-fìlɛ̀mòtʰùʔúbànì	I gave the person Ubani.	Epenthesis - Glottal Stop	L-L	L-L

D Tone Analysis

Table 15: **Tone Analysis in Verbs**

S.no	Underlying Form	Surface Form	Translation	V1-V2 Tone	Surface Tone
1.	χò-àpàjà	χ^{w}àpàjà	To cook.	L-L	L
2.	kè-àpàjà nàmà	k^{j}àpàjà nàmà	I cook meat.	L-L	L
3.	rɛ̀-àpàjà nàmà	r^{j}àpàjà nàmà ràpàjà nàmà	We cook meat.	L-L	L
4.	bá-àpàjà nàmà	bápájá nàmà	They cook meat.	H-L	H (HTS)
5.	ò-àpàjà nàmà	wàpàjà nàmà	You cook meat.	L-L	L
6.	ó-àpàjà nàmà	wápájá nàmà	S/he cooks meat.	H-L	H (HTS)

Table 16: **Tone Analysis in Nouns**

S.no	Underlying Form	Surface Form	Translation	V1-V2 Tone	Surface Tone
1.	mà-ànò	mànò	Plans	L-L	L
2.	bá-áχì	báχì	Builders	H-H	H
3.	mà-úbà	mábà	Droughts	L-H	H
4.	mò-áxì	m^{w}áxì	Builder	L-H	H
5.	bá-èmìsì	báemìsì	Bridesmaids	H-L	H
6.	bá-ètsì	báetsì	Imitators	H-L	H